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Multi-YOLOv8: An infrared moving small object detection model based on YOLOv8 for air vehicle

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ABSTRACT

The detection of infrared moving small objects faces significant challenges in the field of object detection for air vehicles. These types of objects usually occupy a small number of pixels in an infrared image, resulting in limited feature information, considerable feature loss, low recognition accuracy, and various challenges in single-frame detection. To address these challenges, this paper proposes an efficient multi-input method named Multi-YOLOv8, which is based on the YOLOv8s model. The proposed method uses current frames as a primary input and incorporates optical flow processing images and background suppression images as auxiliary inputs to improve detection performance. In addition, an improved method is developed for optical flow computations, named the pyramidal weight-momentum Horn–Schunck (PWMHS) method, which can process optical flows efficiently and precisely. An improved version of the Wise-IoU (WIoU) v3, referred to as α*-WIoU v3, is proposed as a bounding box regression (BBR) loss function to optimize the YOLOv8 network. Further, the BiFormer module and lightweight convolution GSConv are introduced to improve the attention to key information for the objects and balance the computational cost and detection performance, respectively. Moreover, a small object detection layer is added the YOLOv8 network to improve the capability for small object detection. Finally, a warming-up training method that can reduce the dependency on auxiliary inputs and ensure model stability in case of auxiliary input failures is developed. The results of the comprehensive experiments on an open-access dataset reveal that the proposed model outperforms the mainstream models in overall performance. The proposed method can significantly enhance the detection ability of infrared moving small objects.

1. Introduction

Infrared imaging guidance for air vehicles requires research of detection techniques for infrared moving objects in complex backgrounds to achieve high-performance autonomous recognition capabilities. Objects within an infrared field of view are typically characterized by their large distance and occupation of fewer pixels [1]. According to the definition given by the Society of Photo-Optical Instrumentation Engineers, objects with a size smaller than 0.12 % of the total image size are classified as small objects [2]. These objects are characterized by small size, low brightness, blurry shapes, and fast movement speeds compared to a complex background. Considering the current challenges in the field of air vehicles, this study focuses on achieving efficient and accurate detection of infrared moving small objects.

Currently, infrared moving small object detection methods can be broadly categorized into two major classes: single-frame detection methods and multi-frame detection methods. Single-frame detection methods can be further subdivided into three primary categories: morphology-based methods [3], filter-based methods [4–6], and outlier-based methods [7–9]. In filter-based methods, using a hybrid driving mechanism for filter design can effectively balance model performance and network resources [10]. In contrast, the morphology-based and filter-based methods are computationally efficient, but it is challenging to select suitable templates or filters for specific scenarios. The outlier-based methods still have a high false alarm rate in the presence of point-like background features and noise interference.

Therefore, multi-frame detection methods, which use multiple frames of images for object detection and extract spatiotemporal motion information from consecutive frames, have been extensively researched

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[11–13]. However, as demonstrated in [12,13], multi-frame detection algorithms heavily rely on image feature extraction. Namely, when the results of image registrations and optical flow vector calculations fail to meet expectations or when an object is stationary, actual objects could be erroneously considered part of the background. To prevent false positives and ensure model stability, this study focuses on a multi-frame feature extraction method and a reasonable input strategy.

Selecting an appropriate object detection algorithm for a particular scenario is crucial for the rational processing of input data. The You Only Look Once (YOLO) model is a convolutional neural network (CNN)-based object detection model [14], which has evolved into a fundamental real-time object detection model in the fields of robotics, autonomous vehicles, and video surveillance [15–17]. Currently, the YOLOv8 model is the dominant object detection algorithm among the YOLO models due to its significant advantages, such as speed, simplicity, and multi-scale detection. However, the accuracy of the YOLO model's detection is influenced by the quality of the input images. This can limit its ability to recognize small objects, which may lead to detection failures, as demonstrated in Fig. 1.

The YOLOv8 model might achieve false or missed detection results in infrared object detection due to the characteristics of these objects, such as small pixels, fast movement speed, and complex backgrounds. First, a single input cannot carry motion information, which can lead to degradation in the object detection performance for objects with large motions. Second, in complex backgrounds, this model might not pay sufficient attention to the information on small objects. Third, in complex backgrounds, there is a high level of interference that resembles small objects, which increases the probability of false detections. Moreover, small and dark objects with substantial movement distances make detection prone to omissions. Finally, the features of infrared moving small objects are greatly affected by scene factors such as illumination and temperature, which may cause large feature variations and different detection difficulties within the same scene. Therefore, it is necessary to develop a stable and fast detection method to handle the detection of infrared small objects moving rapidly in complex backgrounds.

In recent years, various improved object detection models have been proposed to address the challenges in infrared moving object detection. To solve the problem of the lack of shape and texture features of an object, a single-shot, multi-box detector (SSD) for infrared small object detection was proposed in [18]. In [19,20], a multi-head attention mechanism was added to the YOLOv5s model to detect small objects, and the CIoU loss function was used. When an object is difficult to detect, the region of the object could be refocused with the assistance of region-specific context [21]. Aiming at reducing the interference of complex background, dynamic background extraction was used as an additional input modality to the detector [22]. When detecting multiple objects, an input image could be divided into multiple small blocks for detection [23]. For infrared objects with fewer pixels, a small object detection layer was added to the YOLOv5 model to achieve more precise detection of small objects [24]. These methods can improve detection performance to a certain extent by increasing attention to small objects or adding more suitable structures for detecting small objects. However, these methods do not effectively consider the fusion of object motion information with original frame information to address the challenges posed by small, dim objects in complex backgrounds.

To address the limitations and shortcomings of the existing research, this study designs an efficient multi-frame, multi-input model for detecting infrared moving small objects. The proposed model uses the current frame as a primary input and incorporates the background suppression image and optical flow processing image as auxiliary inputs to improve the detection performance of infrared moving small objects. The proposed model consists of four key modules: background suppression module, optical flow processing module, multi-input processing module, and multi-input object detection module. Specifically, this study proposes a method for detecting infrared moving small objects in complex scenes using multiple inputs, which enhances both the robustness and detection performance of a model. Further, this study investigates the input composition of multiple inputs and proposes a module that uses the detection confidence to determine the input type. Furthermore, this study improves the detection performance for small objects by enhancing the YOLOv8 network structure, which includes modifications of the loss function, attention module, lightweight convolution, network training method, and small object detection layer. The application of multi-frame detection methods to the field of moving object detection is highly valuable. Information obtained from consecutive frames provides insights into object detection, tracking, localization, and category analysis. Finally, the proposed multi-input object detection model not only can improve detection performance but can also provide a strong foundation for future research.

The main research contributions of this study can be summarized as follows:

1. An innovative detection model named Multi-YOLOv8 that uses the current frame, background suppression image, and optical flow processing image as inputs is proposed for infrared moving small object detection. The model can obtain original feature information, object information, and motion information from the inputs, improving the detection performance. In order to optimize the utilization of information from multiple inputs, a proposed multi-input processing module determines the input composition based on the detection confidence which can effectively fuse the features from multiple inputs and balance the detection performance and real-time performance;
2. The α*-WIoU v3 bounding box regression loss function, which can effectively balance the gradient gain of high- and low-quality samples, is proposed. This improvement enhances the model's localization ability and bounding box regression accuracy significantly and also increases the flexibility of the object detection model in various scenarios;

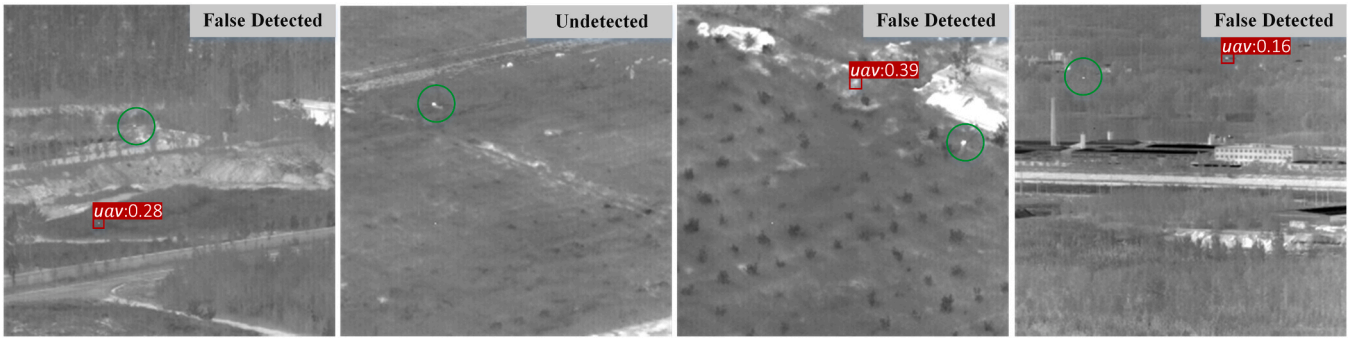


Fig. 1. Examples of YOLOv8 detection failure under complex background. The green circle signifies the object's location, and the red bounding box denotes the results.

3. A warming-up training method is developed to reduce dependency on auxiliary inputs and ensure model stability in case of an auxiliary input failure. Additionally, this study introduces a layer for small object detection and uses the attention mechanism BiFormer to enhance detection capabilities. Moreover, it uses the lightweight convolution GSConv to reduce the model size and maintain detection performance.

The proposed method is evaluated and compared with the existing methods on the Infrared Shimmering Aircraft Detection and Tracking (ISD) dataset [25]. The comparison results show that the proposed method can significantly improve the detection performance in terms of Precision, Recall, AP_{50}, and $AP_{50:95}$ metrics, compared with the existing methods, while maintaining the real-time processing speed.

The structure of the rest of the paper is as follows: Section 2 reviews the related work on multi-frame detection methods, loss function optimization, and YOLOv8 architecture modification. Section 3 provides a detailed introduction to each module of the proposed infrared moving object detection model. Section 4 offers insights into the experimental setup, results, and discussions. In Section 5, the paper is concluded and the main contributions are highlighted.

2. Related work

2.1. Multi-frame and multi-input detection methods

Considering the limitations of single-frame detection methods, researchers have used consecutive frames to extract effective spatiotemporal information to improve the accuracy and efficiency of object recognition. Representatively, in [26], the localized image alignment was used so that consecutive frames in a sequence were accumulated to suppress noise. In [27], the authors segmented the images, performed registration, and applied the stable principal component tracking method to recognize the segmentation results. In [28], the optical flow (OF) method was employed to densely sample the motion feature points of infrared objects, thus effectively detecting infrared small objects with motion. In [29], differences between consecutive images were used to detect large motions by warping the optical flow. An optical-flow-feature fusion-based video object detection method was proposed in [30], considering temporal coherence between video frames. The above-mentioned studies have processed multi-frame images by using background suppression or optical flow methods. Although the input to an object detection network includes motion information, object information of the original frames could be lost. For complex backgrounds, the aforementioned detection models may mistakenly identify backgrounds with significant motion as objects. In [31], image registration was used to extract multi-frame motion information, which was then used as input data alongside the original image for infrared object detection. However, the original frame was fed to the object detection network along with the background-suppressed image, and when the original frame could alone achieve satisfactory detection results, using additional input data might be computationally wasteful. In [32], different input fusion methods were employed to determine the stability of a multi-input model. Nevertheless, the existing methods have not considered using a more reasonable composition of multi-input that could ensure the object detection model maintains excellent detection performance while guaranteeing real-time efficiency. Moreover, the effective fusion of multi-frame motion information and original frame information, as well as the model robustness after the failure of background suppression or optical flow methods, have not been fully considered. To address these shortcomings, this paper proposes a multi-input object detection model whose input includes both motion information and original frame information. The input of the proposed model can be divided into single and multiple inputs. The proposed model can improve the detection performance while saving computational power to the greatest extent.

2.2. Bounding box regression loss function

The loss function for bounding box regression (BBR) [33] takes into account various geometric factors related to the bounding box. The generalized IoU (GIoU) was proposed in [34] by incorporating the concept of a minimal closure zone. The minimal closure zone represents the smallest rectangular box capable of enclosing both the prediction box and the ground-truth box. In [35], the distance -IoU (DIoU) that extended the GIoU was designed by incorporating the distance between the centers of the predicted and ground-truth boxes. In addition, the complete-IoU (CIoU) was developed based on the DIoU, considering the difference in aspect ratios between the predicted and ground-truth boxes. In [36], the authors developed the efficient-IoU (EIoU) by replacing the aspect ratio consistency in the CIoU with a separate calculation of aspect ratio consistency. As the BBR loss function in the YOLOv8, the CIoU might hinder the effective optimization of the model when the predicted and ground truth boxes have the same aspect ratio but different width and height values; also, the computation process of the CIoU is complex. In infrared detection tasks, the size, brightness, and distance of objects exhibit significant variations within the field of view. The clarity and quality of images captured by infrared cameras may vary considerably. In addition, most existing methods assume that the training data are of high quality and, therefore, focus on increasing the fitting ability of the BBR loss. However, indiscriminately reinforcing the BBR for low-quality data samples can reduce the localization performance. These shortcomings could be solved by the Wise-IoU (WIoU) [37] through the allocation of small-mass gradient gains using a focusing mechanism, which allows the BBR to focus on ordinary-quality anchor boxes. The Wise-IoU can be roughly categorized into three types: WIoU v1, which constructs an attention-based bounding box loss, and WIoU v2 and WIoU v3, which add a focus mechanism through the gradient gain method using the WIoU v1 as a basis. The WIoU can be used to improve the accuracy and mAP of the YOLO model [38], but it also has certain limitations in infrared moving object detection, such as lower BBR accuracy and insufficient precision in model detection at high AP. This study applies a power transformation [39] on the WIoU v3 to improve the accuracy of bounding box (bbox) regression by adaptively reweighting the losses and gradients of high- and low-IoU objects.

2.3. Object detection networks

In recent years, significant advancements have been made in the field of object detection using various CNN models. These approaches can be broadly categorized into two structural groups: two-stage methods and end-to-end methods. The former include the Region-CNN (R-CNN) [40], Faster-RCNN [41], and Cascade R-CNN [42], and the latter include the SSD [43], Retinanet [44], YOLO [45], Yolov3 [46], and Fcos [47]. Two-stage methods can achieve higher accuracy than the end-to-end methods but have a slower computational speed. Meanwhile, the end-to-end methods have a computational advantage at the expense of detection performance. The end-to-end methods are suitable to be used in an object detection module of air vehicles.

The YOLO algorithm has undergone numerous and extensive enhancements, especially with the introduction of the YOLOv4 model [48]. In [15], the authors used the SPD-Conv to extract multi-scale features to enhance the feature extraction capability of the model for small objects. In [49], the channel- and spatial-focused modules were introduced to enhance the focus of the YOLOv3 model on key regions. In [50], the convolutional block attention module (CBAM) was introduced into the YOLOv8 model to detect small prohibited items with multiple occlusions. The YOLOv8 is the latest version of the YOLO model family, and it can achieve a remarkable balance between detection accuracy and efficiency. In [51], an innovative attention mechanism called BiFormer was introduced to enhance the performance of the YOLOv8 backbone network and make the model's focus on key information. In [52], the GSConv module was used to design a lighter YOLOv5 network structure,

which provided faster model execution while maintaining detection accuracy. In general, in infrared object detection, the main improvements to the YOLO models mainly refer to the application of attention mechanisms, detection structures for small objects, and appropriate convolution and loss functions. In this way, the discriminability of infrared small objects and backgrounds and model attention on moving objects can be significantly improved.

3. Methods

The workflow diagram of the proposed infrared moving object detection model named Multi-YOLOv8 is shown in Fig. 2. The proposed model uses the original frame as a primary input while combining the background suppression and optical flow processing images as auxiliary inputs to improve the detection performance for infrared moving small objects. The original frame, serving as a primary input for detection, contains a substantial amount of object and background information, making the detection process prone to failures in complex backgrounds. The background suppression images and the optical flow processing images are used as auxiliary inputs to provide motion and object information, and they are combined with the main input to accomplish the detection in complex scenarios. The purpose of the two auxiliary inputs is to assist the primary input sequentially in completing the detection task in scenarios with varying difficulty. In addition, if one of the auxiliary inputs fails, the model can still maintain robust detection performance.

First, in the multi-frame processing phase, the background suppression module aligns the current frame with the previous frame and performs the background suppression to generate a background suppression image. The optical flow processing module uses the improved Horn–Schunck (HS) method to process the current and previous frames to obtain an optical flow image.

Then, the multi-input processing module decides whether to enable the background suppression module and optical flow processing module for the next frame based on the detection confidence value of the output of the previous and current frames. Based on the decision, the multi-input processing module processes the current frame, background suppression image, and optical flow processing image separately. This processing method enables the thorough integration and effective utilization of information obtained from multiple inputs. After processing, the results are sent to the object detection module.

Finally, the multi-input object detection module detects objects, and the obtained detection confidence value is used as an input for the next frame. The baseline model in this study is the YOLOv8 model, which is optimized in terms of the loss functions, attention mechanism, lightweight convolutions, and small object detection layer. Individual modules will be discussed in subsequent sections.

3.1. Improved optical flow method

3.1.1. Optical flow method based on gradient

Based on the fundamental assumption of image grayscale constancy between consecutive frames, gradient-based methods [53] estimate the optical flow using image grayscale gradients and obtain optical flow constraint equation (Eq. 1):

$$I_x u(x,y) + I_y v(x,y) + I_t = 0 \tag{1}$$

where, $I(x,y,t)$ is the gray value at moment t of point m. $u(x,y)$ and v (x, y) are the x and y components of optical flow at position(x,y).

The Horn-Schunck algorithm proposes a smoothness constraint on the optical flow, as shown in Eq. 2:

$$E_s = \int\int \left(u_x^2 + u_y^2 + v_x^2 + v_y^2\right) dxdy \tag{2}$$

The equation can be solved by using the Gauss-Seidel method, Eq. 3 and Eq. 4 form the basic equation of the Horn-Schunck method, the expressions are given below:

$$u^{(k+1)} = \overline{u}^{(k)} - I_x\left(I_x\overline{u}^{(k)} + I_y\overline{v}^{(k)} + I_t\right) \Big/ \left(\lambda^2 + I_x^2 + I_y^2\right) \tag{3}$$

$$v^{(k+1)} = \overline{v}^{(k)} - I_y\left(I_x\overline{u}^{(k)} + I_y\overline{v}^{(k)} + I_t\right) \Big/ \left(\lambda^2 + I_x^2 + I_y^2\right) \tag{4}$$

where, I_x, I_y and I_t represent the partial derivatives of the image grayscale with respect to x, y and t, respectively. $u^{(0)}$ and $v^{(0)}$ denote initial velocity estimates of optical flow. $(\overline{u},\overline{v})$ is the local average. λ is the

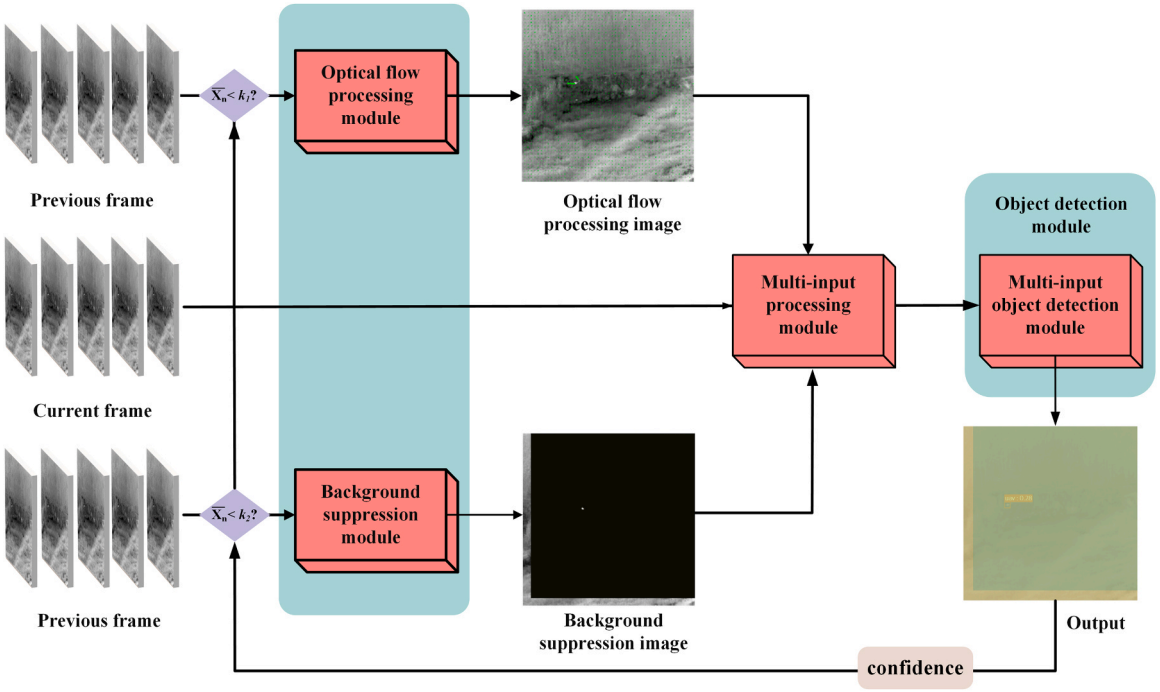


Fig. 2. The workflow of the Multi-YOLOv8 model.

weighting parameter. k is the number of cycles.

3.1.2. Pyramidal Weight-Momentum Horn-Schunck (PWMHS) method

The optical flow calculated by the HS method can be unreasonable at certain points. The pixels of infrared small objects are very few in the context of an air vehicle's field of view, and the contours of the moving object area are relatively blurry. The HS method requires a certain number of iterations to meet accuracy requirements, which not only reduces computational efficiency but also increases interference from complex backgrounds. This study improves the iterative equation of the Horn–Schunck method and introduces a pyramid computation method. The improved method is named the pyramidal weight-momentum Horn–Schunck (PWMHS) method. First, to reduce the convergence time and ensure a highly stable optical flow field, this study employs the method of adding a momentum term [54] to speed up the computation of the optical flow field. Second, a weight function in Eqs. 5 and 6 suppress the optical flow computation of background pixels, which reduces the overall computational burden. Finally, a pyramidal hierarchical computational approach is introduced, which allows the PWMHS method to achieve high accuracy with fewer iterations, thus improving the real-time performance and detection capabilities.

$$\omega_1\left(x,y\right)\begin{cases}1 & I_x^2+I_y^2>T\\ \left(\left(I_x^2+I_y^2\right)/T\right)^{\alpha_1} & I_x^2+I_y^2<T\end{cases} \tag{5}$$

$$\omega_2\left(x,y\right)\begin{cases}1 & I_x^2+I_y^2>T\\ \left(\left(I_x^2+I_y^2\right)/T\right)^{\alpha_2} & I_x^2+I_y^2<T\end{cases} \tag{6}$$

In Eqs. 5 and 6, α_1 suppresses the magnitude of the optical flow vector, and α_2 regulates the ratio of the fundamental equation constraint to the smoothness constraint. The parameter T is the threshold of the introduced weighting equation.

A momentum term and a weight function $\omega(x,y)$ are added to Eqs. 3 and 4, and the following equations can be obtained.

$$u^{(k+1)}=\omega_1\left(x,y\right)\overline{u}^{(k)}-I_x\frac{I_x\overline{u}^{(k)}+I_y\overline{v}^{(k)}+I_t}{\lambda^2+I_x^2+I_y^2}\cdot\omega_2\left(x,y\right)+\rho\cdot\left(u^k-u^{(k-1)}\right) \tag{7}$$

$$v^{(k+1)}=\omega_1\left(x,y\right)\overline{v}^{(k)}-I_y\frac{I_x\overline{u}^{(k)}+I_y\overline{v}^{(k)}+I_t}{\lambda^2+I_x^2+I_y^2}\cdot\omega_2\left(x,y\right)+\rho\cdot\left(v^k-v^{(k-1)}\right) \tag{8}$$

The range of$\rho$is between 0 and 1.

This study assumes that $\alpha_1=\alpha_2=\alpha$ based on numerous experiments. To further improve the efficiency of the model operation, $\omega(x,y)$can be replaced by a piecewise linear function.

Fig. 3(a) shows how the graph of $f(x)=x^{\alpha}$ varies with the value of α; Fig. 3(b) shows the approximation relationship between the piecewise linear function and the original function$\omega(x,y)$under different parameters k'.$\omega(x,y)$ can be approximated when the value of α is greater than 20.

$$\omega\left(x,y\right)\begin{cases}1 & I_x^2+I_y^2\geq T\\ \left(1/\left(1-k'\right)\right)\cdot\left(I_x^2+I_y^2/T\right)-k'/\left(1-k'\right) & k'T<I_x^2+I_y^2<T\\ 0 & I_x^2+I_y^2\leq k'T\end{cases} \tag{9}$$

In Eqs. 7–9, setting different values of parameter k' enables the exclusion of certain pixels from the iterative calculation of the Horn–-Schunck method and the suppression of their optical flow vectors. This is because the speed of moving objects is relatively higher compared to the background moving speed.

In infrared moving object detection, the stability of an optical flow method is influenced by the relative velocity and distance between an object and a camera. Therefore, aiming to enhance the stability of the optical flow method, this study employs the pyramid computation method. The pyramid computation method is a multi-scale analysis technique that has been proven to be especially effective in handling motion information at various scales, particularly when an object is situated at a considerable distance from or in proximity to the camera. This method helps to maintain the global consistency of the optical flow

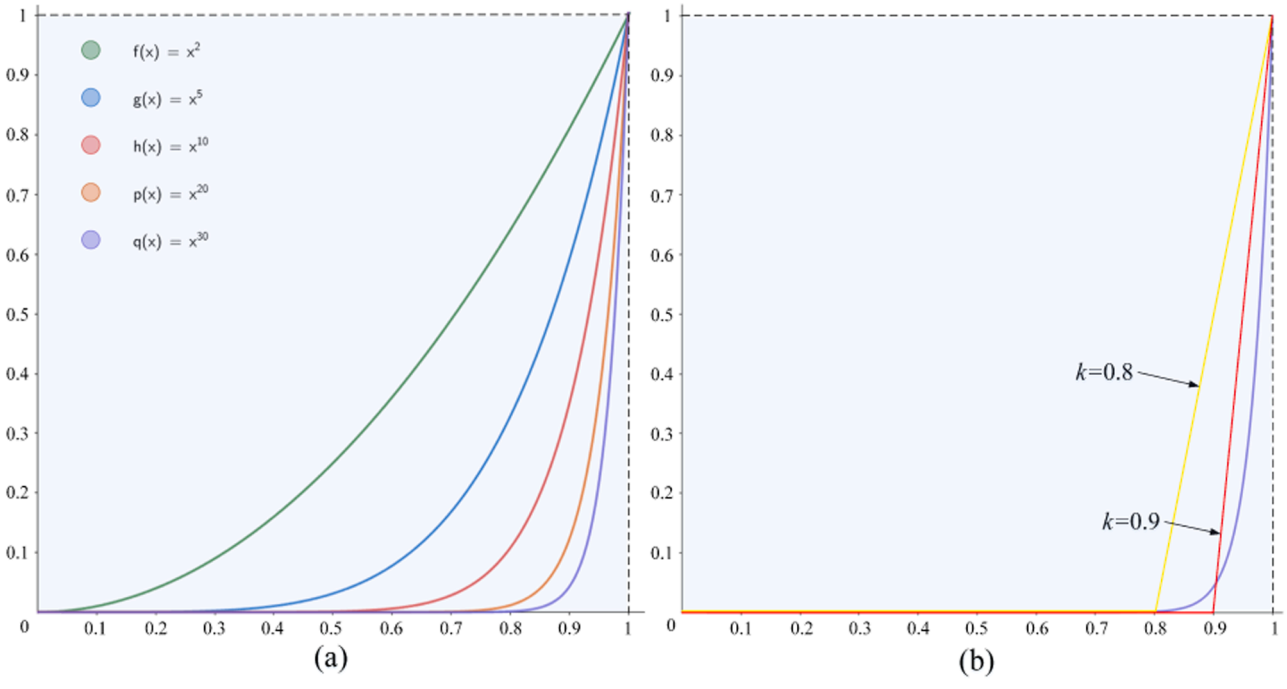


Fig. 3. (a) Graph of functions$f(x)=x^{\alpha}$; (b) Graph of the piecewise linear functions for different values of k'.

field, allowing the algorithm to detect objects robustly in an image sequence. The three-layer image pyramidal structure is presented in Fig. 4.

The workflow of the PWMHS method is shown in Fig. 5, where it can be seen that in the first step, the current and previous frames are downsampled to the second layer using an image pyramid, and the PWMHS method is applied to calculate the optical flow between these two frames. In the second step, the previous frame is translated in the first layer by a specific distance, which is determined by the optical flow values calculated at the second layer, and the optical flow between the translated and current frames is computed to produce the motion vectors. Finally, the calculated optical flows of the first and second layers are combined through vector addition. This process is repeated until the topmost layer, which corresponds to the original image, is reached. The PWMHS method adopts a progressive iterative strategy to improve the optical flow estimation from low to high resolution, reducing the computational and storage costs while eliminating the image details and noise interference and enhancing image smoothness and consistency. Moreover, the PWMHS method can enlarge the optical flow search range, cope with large-scale pixel motion, and maintain the relative size of the integration window, improving the accuracy of the optical flow.

More specifically, the PWMHS method calculates the optical flow through the following steps. For a point (x, y), d^L is the optical flow calculated of the Lth layer, g^L is the initial optical flow for the Lth layer, and v^L is the calculated value by the optical flow method; $g^L = 0$, and $d^L = v^L + g^L$. After multiplying the d^L value by two and upsampling operation, the initial motion vector of layer $(L - 1)$ is $g^{L-1}=2*d^L$. The previous frame of the Lth layer is translated $gL-1$; $v^{L-1}$is calculated from the previous image after translation along with the current frame at $(L - 1)$ layer. The above steps are repeated to obtain the final result.

For the PWMHS method, the following assumptions regarding the hyperparameter settings are adopted: α and k' are are the weighting function's parameters. This study recommends setting the value of parameter α to be in the range of [20,80], where a larger α value reduces the number of iterative computations. The parameter k' can be adjusted based on particular experimental needs, and suggested combinations for α and k' are (20, 0.8) or (30, 0.9). The general setting of the T value depends on the spatial variation of the pixel values, illuminance changes, and a specific scenario. For scenarios with large object motion or objects lingering near image edges, a range of [15,30] is recommended. For scenarios with small object motion, a range of [5,15] is recommended. The momentum constant ρ should be in the range [0,1],

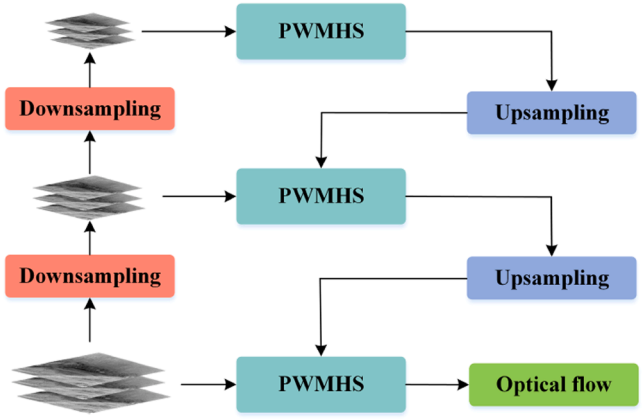


Fig. 5. Workflow of the PWMHS method. The downsampling process uses a Gaussian 5×5 low-pass filter to convolve the image. This process samples the result in spaced rows and spaced columns to generate the image for the lower layer. The upsampling interpolates zeros into the image and generates the upper layer by convolving with the filter and multiplying by four.

and it is recommended that it does not exceed 0.8 to achieve a stable optical flow field and reduce the number of iterations. The optical flow constant λ controls the smoothness of the optical flow field; a larger λ results in a smoother flow field, whereas a smaller λ value focuses on more detailed changes. The selection of the λ value is influenced by the characteristics of data and an application scenario. For scenarios with complex backgrounds, high noise levels, small objects, large object motion, low image resolution, and low brightness, the value of λ should be in the range [50,500]. The value of λ can be tuned from zero for different datasets. Based on many experimental results, the optimal number of iterations is in the range of [10,30] or fixed at 16.

3.2. Background suppression method

The background suppression results depend on the image registration. The main principle of the homography estimation [55] is that two cameras placed different positions take pictures of the same object, and the captured frames are represented by a homography matrix $\mathbf{H_{AB}}$ to represent the transformation relation.

In the scenario with infrared moving objects, an object can be

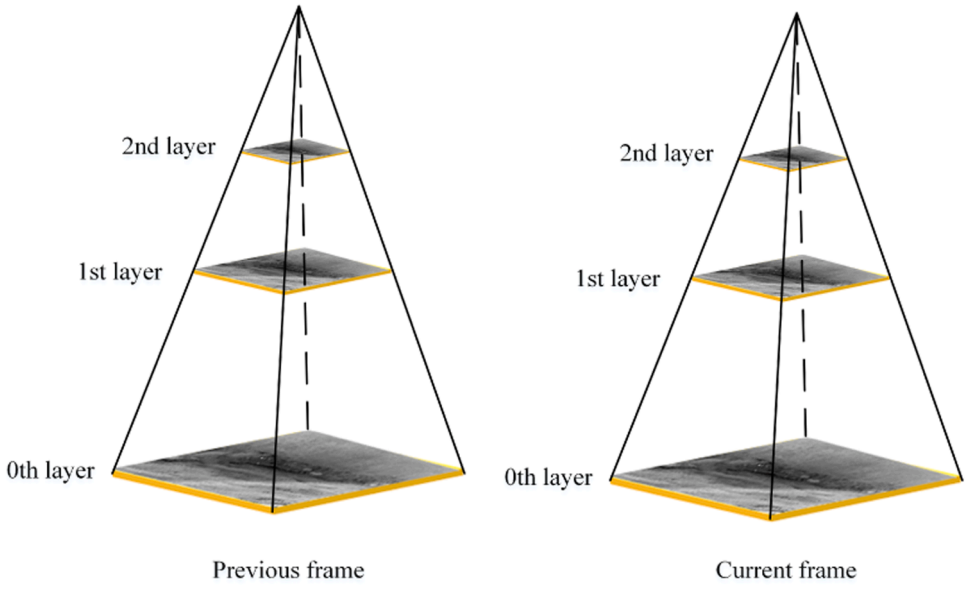


Fig. 4. The structure of the 3-layer image pyramid.

considered stationary while the detection device undergoes rigid-body motion relative to the object; there is homography between consecutive frames (Fig. 6). Due to changes in the camera position relative to an object, the object shape can vary between two frames. Nevertheless, the transformation between pixel coordinates corresponding to the same object in these frames can be characterized by the homography matrix **H** (Eq. 10), where $(x_A, y_A)$and $(x_B, y_B)$are coordinates of pixels in the left image and the right image, respectively. In the scenario of infrared moving aircraft detection, the infrared camera is situated at a considerable distance from the aircraft. Compared with the long distance between camera and aircrafts, the moving of the camera can be ignored. Consequently, the transformation between two images can be simplified to a Euclidean transformation (Eq. 11), which encompasses translation and rotation. Under normal circumstances, the camera does not have rotational motion in the field of view, so the transformation between two images can be reduced to a translation transformation (Eq. 12).

$$\begin{pmatrix} x_B \\ y_B \\ 1 \end{pmatrix} = \boldsymbol{H} \begin{pmatrix} x_A \\ y_A \\ 1 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \begin{pmatrix} x_A \\ y_A \\ 1 \end{pmatrix} \tag{10}$$

$$\boldsymbol{H} \approx \begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{pmatrix} \tag{11}$$

$$\boldsymbol{H} \approx \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \tag{12}$$

In Eq. 12, t_x and t_y represent the pixels distance of translation in x and y directions. This paper uses the fast homography estimation net proposed in [31] for parameter estimation of the homography matrix **H**, and its working process is illustrated in Fig. 7. The structures of the feature extraction module and the parameter estimation module are shown in Fig. 8.

The terms in Fig. 8 is explained in Table 1.

After obtaining the homography matrix, the previous frame is transformed, and a registration image is generated. The background suppression image is obtained by subtracting the registration image from the current frame, as shown in Fig. 9.

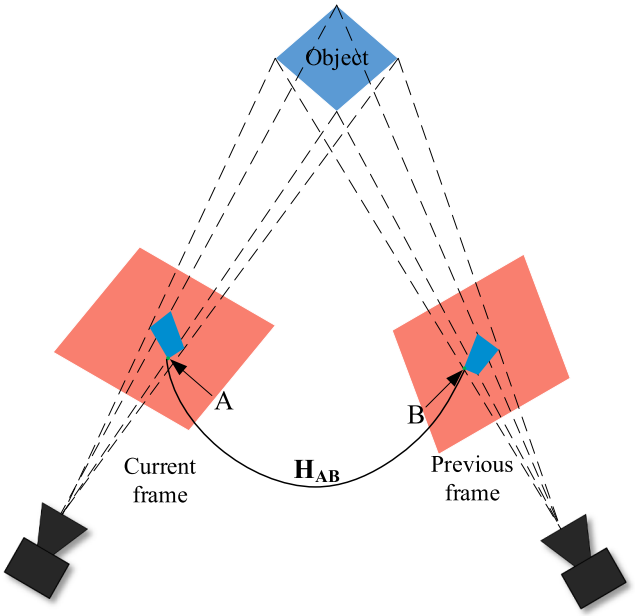


Fig. 6. Homography between two adjacent frames.

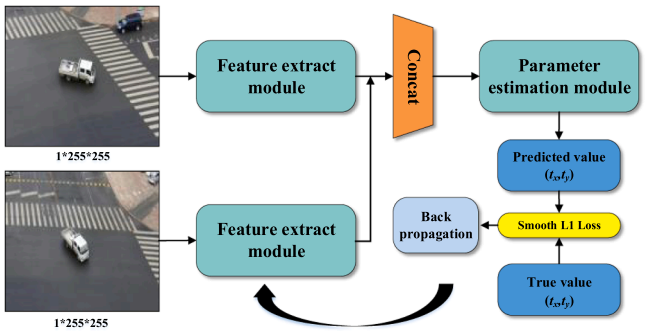


Fig. 7. The structure of fast homography estimation network.

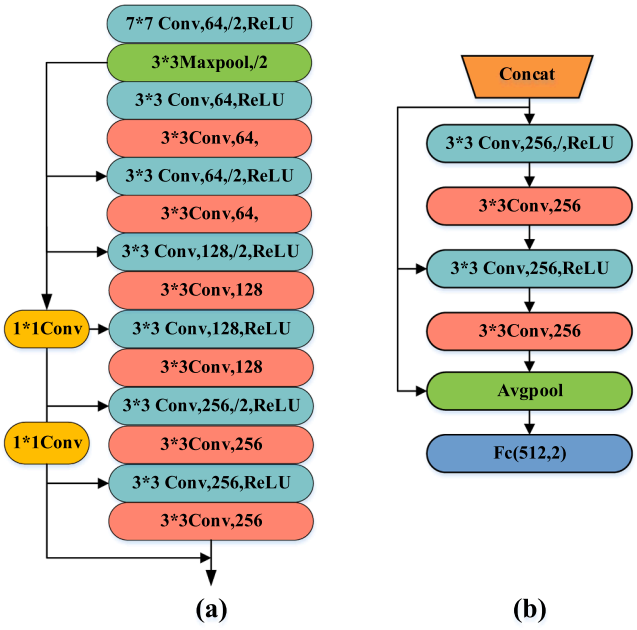


Fig. 8. (a) The structure of the feature extract module; (b) The structure of the parameter estimation module.

Table 1
Detailed explanation of individual modules in Fig. 8.

Noun	Explanation
7*7 Conv, 64, /2	A convolutional layer with 64 filters of size 7×7, a stride of 2, with a Batch Normalization.
ReLU	Rectified Linear Unit.
Maxpool	Max pooling layer.
Avgpool	Average pooling layer
Concat	The inputs are stacked in the channel direction.
Fc (512,2)	A fully connected layer is utilized with 512 input units and 2 output units.

3.3. Multi-input processing module

The object detection network used in this study is designed based on the YOLOv8s model, and multiple inputs are generally input into the network model in two ways: thorough superimposition in the RGB channel and thorough separate processing. Based on the characteristics of the RGB channel, the three inputs are fed to the RGB channel for superposition. Particularly, the current frame is input in the B channel, the optical flow processing image is input in the G channel, and the background suppression image is input in the R channel. After merging these three images, a focus module is employed for the extraction and

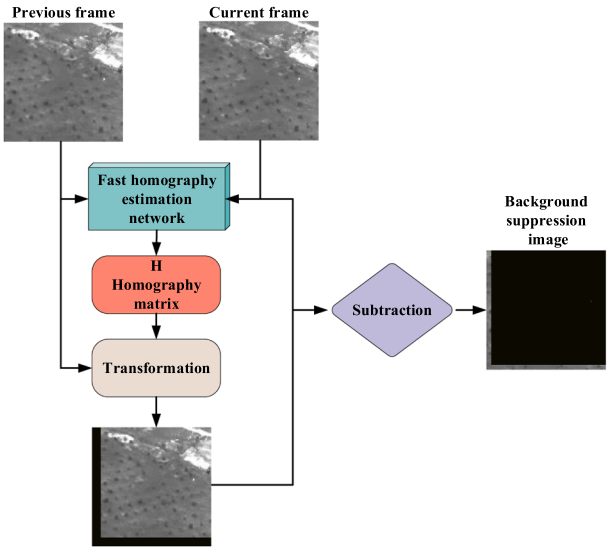


Fig. 9. The workflow of the background suppression module. A fast homography estimation network estimates the homography matrix between two consecutive frames. The homography matrix transforms the previous frame, resulting in image registration. Pixel-wise subtraction on the two images achieves background suppression, producing the final background suppression image.

fusion of features of the RGB three-channel image.

However, multiple inputs have different image characteristics. The primary input is the current frame, which serves as the original image. This input contains a large number of pixels necessary for object localization but also noisy pixels and a large number of complex background pixels. The auxiliary inputs consist of background suppression images and optical flow processing images. The background suppression image is reduced by most of the background pixels, and the portion of the optical flow processing image with the larger optical flow vector represents the position of an object. Due to the differences in the characteristics of the primary and auxiliary inputs, features of multiple input images need to be extracted separately using different feature extraction methods to ensure that information can be fully explored and used. The workflow diagrams for the two types of image processing are shown Fig. 10.

The background suppression module and the optical flow processing module introduce additional time costs. Therefore, to achieve an optimal balance between accuracy and real-time performance, the proposed method inputs the detection confidence values of the previous and current frames into the multi-input processing module as feedback. This feedback is used as a basis for deciding whether to turn on the background suppression module and the optical flow processing module in the next frame or not. In Fig. 10, $\overline{X}$and $\overline{X_{n-1}}$ represent the detection confidence values of the current and previous frames' outputs, respectively. The average values are compared with the corresponding predefined detection confidence threshold values, which are set to 0.35 and 0.5 in this study, and the comparison result determines whether the background suppression module and the optical flow processing module will be activated or not. It should be noted that in this study, the model uses only the original frame as input for detecting the first frame and sets$\overline{X_{n-1}}$to 0.5. To avoid constant switching between single- and multi-input, the proposed method incorporates the confidence value of the previous frame output, enhancing the model's stability. This approach not only ensures detection accuracy but also conserves computational resources, thus reducing the time required for the model detection process.

By comparing the detection confidence with the predefined threshold values, the input composition is obtained. Three Focus (in, out) modules are adopted to extract features from three images, where "in" and "out" denote the numbers of input and output channels, respectively. The Focus module is a special downsampling module that splits the input image into four subsampled images. Then, an average pooling layer with a 5×5 kernel size is applied after the Focus module to filter out the noise in the features of the background suppression image and the optical flow processing image. A max pooling layer with a 5×5 kernel size is also added after the Focus module to improve the features of the object region. Finally, the features of the original frame, the background suppression image, and the optical flow processing image are fused together by the "Concat" operation and the Conv module, and the features of multiple inputs are extracted and enhanced. This study uses the multi-input processing module to replace the origin input layer of the Yolov8s model.

In general, the multi-input processing module first determines the number and type of inputs based on the detection confidence value, which ensures detection accuracy and real-time performance. Then, the module performs the distinct feature processing for different input types, ensuring the full utilization of the original frame and motion

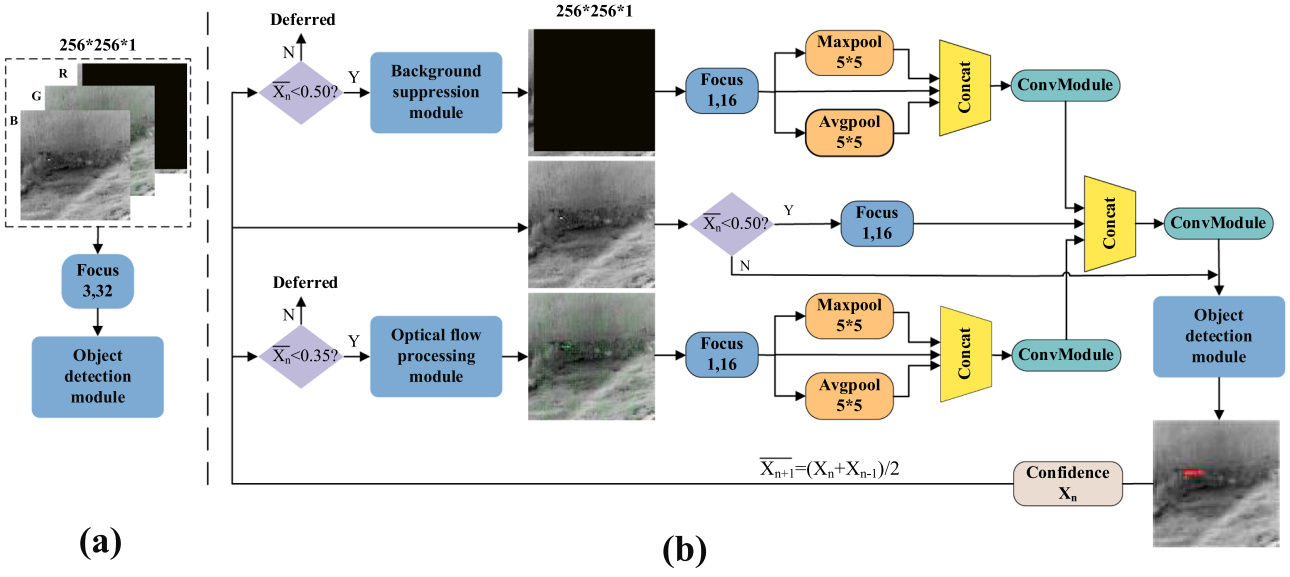


Fig. 10. (a) The multiple inputs are placed into the RGB channels; (b) The multi-input processing module proposed in this study.

information. Finally, the detection confidence value of the object detection module's output is fed back to the multi-input processing module as input for the next frame. This module allows the Multi-YOLOv8 model to form a closed loop, and the input composition depends on the confidence of detection. In simple scenarios, the Multi-YOLOv8 model operates with a single input to ensure real-time detection, and in complex scenarios, the model operates with multiple inputs to ensure detection performance.

This study recommends setting different detection confidence thresholds for different detection task requirements and application scenarios. The recommended combination of confidence thresholds based on extensive experiments are (0.25, 0.45), (0.35, 0.5), and (0.45, 0.55). The positions of the background suppression module and optical flow processing module can be changed according to a particular application scenario.

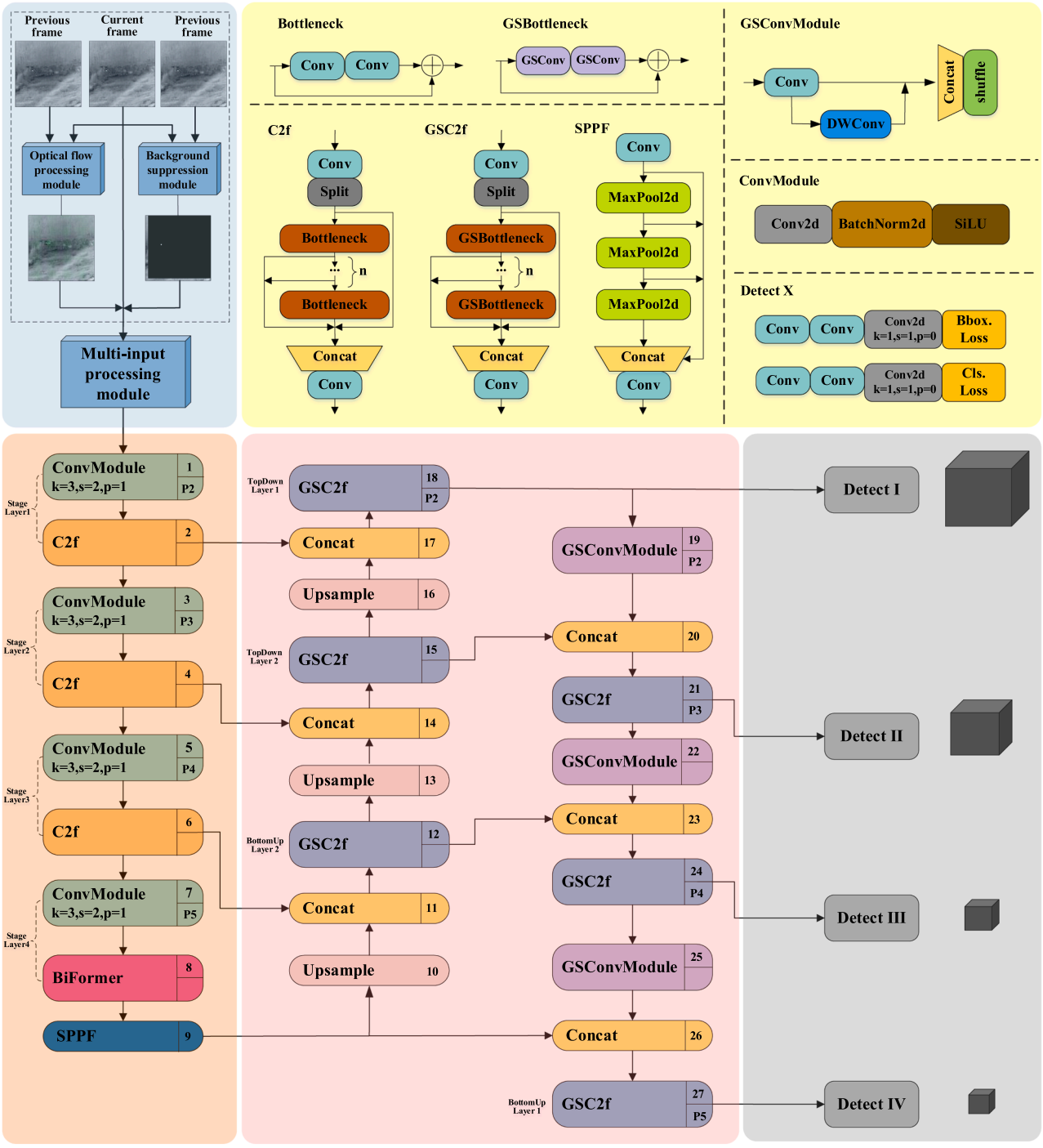


Fig. 11. The Multi-YOLOv8 model structure. The YOLOv8s is selected as the baseline model in this study. The current and previous frames are processed separately using the optical flow processing and background suppression modules. The processed results, along with the original frame, are input into the multi-input processing module. In this module, the feature maps undergo further processing before being forwarded to the YOLOv8-based object detection model, which includes four detection scales. The input to the object detection module is influenced by the detection confidence of the previous two frames during the detection process but is unaffected during the training process.

3.4. Multi-YOLOv8 model

This section presents the Multi-YOLOv8 model, which is an air vehicle infrared moving small object detection model based on the YOLOv8s model. The Multi-YOLOv8 model processes the optical flow, background suppression, and current frames using the multi-input processing module and feds them to the YOLOv8s-based object detection module. The Multi-YOLOv8 model network is presented in Fig. 11.

Firstly, α*-WIoU v3 is proposes as a new loss function that balances bounding box regression accuracy for high- and low-quality samples. Function α*-WIoU v3 has an adjustable parameter, α*, which provides flexibility in terms of regression accuracy; also, the method has smaller regression errors compared to using the CIoU function. Second, this study improves the YOLOv8 model by introducing the BiFormer module [56], which uses a dynamic sparse attention mechanism with the Bi-level Routing Attention (BRA). The BRA provides the model with higher flexibility in allocating computational resources. The attention mechanism in the BiFormer enhances the model's ability to focus on critical information, thus optimizing the detection performance while reducing computational complexity. The YOLOv8 model substitutes original Conv modules with advanced lightweight GSConv modules, which reduces the computational cost without compromising accuracy. In addition, a small object detection layer is added to the YOLOv8 model, which increases the number of detection scales to four and improves the model's ability to detect small objects. Finally, a warming-up training method is used in the model training process to avoid the excessive influence of the auxiliary inputs on the detection model's performance while maintaining high detection accuracy. This training method optimizes the use of auxiliary inputs to achieve object detection while preserving the impact of the primary input.

3.4.1. Improved loss function

In air-to-air object detection for air vehicles, objects typically occupy a relatively small proportion of the overall pixels. Therefore, a well-designed loss function is crucial for improving the model detection accuracy and avoiding instances of missed or false detection. The YOLOv8 model uses the distribution focal loss and CIoU to compute the bounding box regression loss. However, the CIoU has certain shortcomings. First, the CIoU does not consider the balance of different samples. Second, the CIoU introduces the aspect ratio as a penalty factor in the loss function. Nevertheless, if the real and predicted boxes share the same aspect ratio but differ in width and height values, the penalty term fails to differentiate accurately between these two boxes. Third, the CIoU formula involves the computation of an inverse trigonometric function, which increases the overall computational cost of the model. The CIoU formula is expressed as follows:

$$\mathscr{L}_{CIoU} = 1 - IoU + \rho^2\left(b, b^{gt}\right) \Big/ \left(W_g^2 + H_g^2\right) + \alpha v \tag{13}$$

where IoU denotes Intersection over Union, $\rho^2(b, b^{gt})$ denotes the Euclidean distance between the centroids of the real box and the prediction box. W_g and H_g denote the width and height of the minimum enclosing box formed by the real and predicted boxes. The expressions for α and v in Eqs. 14 and 15 are as follows:

$$v = 4*(arctan(w^{gt}/h^{gt}) - arctan(w/h))^2 \Big/ \pi^2 \tag{14}$$

$$\alpha = v/(1 - IoU + v) \tag{15}$$

where h^{gt} and w^{gt} denote the height and width of the real box.

Unlike the fixed-focus mechanism used by many of the traditional loss functions mentioned earlier, the WIoU [37] considers the aspect ratio, centroid distance, and overlap area while incorporating a dynamic non-monotonic focusing mechanism to address the challenge of balancing between better and worse quality samples in the BBR. Currently, there are three versions of the WIoU models, and the WIoU v3 is selected as a baseline version in this study. The computational equation of the WIoU v3 is given in Eq. 16:

$$\mathscr{L}_{WIoUv3} = r \times \mathscr{L}_{WIoUv1} \tag{16}$$

Where the elements in Eq. 16 are calculated by Eqs. 17–20.

$$\mathscr{L}_{WIoUv1} = R_{WIoU} \times (1 - IoU) \tag{17}$$

$$R_{WIoU} = \exp\left(\frac{\left(b_{C_x}^{gt} - b_{C_x}\right)^2 + \left(b_{C_y}^{gt} - b_{C_y}\right)^2}{\left(W_g^2 + H_g^2\right)}\right) \tag{18}$$

$$r = \beta/\delta\alpha^{\beta-\delta} \tag{19}$$

$$\beta = \mathscr{L}^*_{IoU}/\overline{\mathscr{L}_{IoU}} \in [0, +\infty) \tag{20}$$

In these equations , $(b_{b_x}, b_{b_y})$and$(b_{c_x}^{gt}, b_{c_y}^{gt})$ denote the coordinates of the center points of the real box and the prediction box; β is the outlier that measures the quality of the anchor box, and a non-monotonic focus factor r is constructed based on β; $\delta$and$\alpha$are hyperparameters that can be adjusted to fit different models; $\mathscr{L}^*_{IoU}$ is the constructed monotonic focusing factor coefficient; $\overline{\mathscr{L}_{IoU}}$ is the running mean with momentum m.

In the WIoU v3 loss function, the quality of an anchor box is inversely proportional to the value of β, which defines the value of r assigned to it, thus affecting the weight of high-quality anchor frames in the total loss function. A large value of β implies a low-quality anchor box, which yields a small gradient gain and a reduced harmful gradient.

However, during the detection of infrared moving small objects, the proportion of pixels occupied by an object in the field of view is influenced by factors like relative distance and object type. Also, the recognition difficulty of the same object might vary over time due to the complex background. Therefore, this study aims to enhance the robustness of the detection model to adapt to different scenes and achieve more generalized detection performance. In addition, due to the characteristics of the object, such as small, dark, and feature ambiguity, achieving accurate bounding box regression is crucial for obtaining precise information on the position and size of an object.

In pursuit of a more balanced positioning and sizing of object bounding boxes, the enhancement of model generalization across various object scenarios, and effective deployment across diverse detection environments, this study introduces a parameter α* into the WIoU v3, deriving an improved loss function denoted by α*-WIoU v3. This parameter is designed to improve regression accuracy without increasing the training or inference time. In this way, the model can address the challenges of infrared moving small object detection better, maintaining adaptability to different scenarios and ensuring efficient training and inference. The expression of α*-WIoU v3 loss function is as follows:

$$\mathscr{L}_{\alpha^*-WIoUv3} = (r \times R_{WIoU})^{\alpha^1} - (r \times R_{WIoU} \times IoU)^{\alpha^2} \tag{21}$$

In Eq. 21, hyperparameters α^1 and α^2 are introduced for different product equations in the α*-WIoU v3, but based on a large number of experimental results, $\alpha^1 = \alpha^2$ is finally selected, and the parameters are denoted as α*. The final loss function is defined by Eq. 22:

$$\mathscr{L}_{\alpha^*-WIoUv3} = (r \times R_{WIoU})^{\alpha^*} - (r \times R_{WIoU} \times IoU)^{\alpha^*} \tag{22}$$

Specifically, the existing WIoU series are transformed to the power form with a single power parameter α*. The loss and gradient of different IoU objects can be adjusted through the adjustment of parameter α*. Based on Eq. 22, $(r \times R_{WIoU})^{\alpha^*}$ can be treated as a constant term k during each loss computation. By moving k to the left side of the equation, Eq. 22 can be regarded as $\mathscr{L}_{\alpha^*-\mathrm{IoU}} = 1 - IoU^{\alpha^*}$. The power transformation in $\mathscr{L}_{\alpha^*-\mathrm{IoU}}$ preserves key properties of $\mathscr{L}_{\mathrm{IoU}}$ as a performance metric, including non-negativity, identity of indiscernible, symmetry, and triangle inequality [39]. In addition, α*-IoU losses have the

properties of relative loss reweighting and relative gradient reweighting. Relative loss reweighting property indicates that $\mathscr{L}_{\alpha^*-\mathrm{IoU}}$ can adaptively down- and up-weight the relative loss of all objects according to their IoUs values when $0 < \alpha^* < 1$ and $\alpha^* > 1$, respectively. When $\alpha^* > 1$, the model focuses on high IoU objects to improve both the localization and detection performance, which is because the model has a stronger ability to predict high IoU objects and the model is more accurate under high AP. When $\alpha^* \in [2,3]$, the model accelerates the learning of positive IoU objects during training at AP_{50} and performs more competitively in terms of loss calculation than those with $\alpha^* \in [1,2]$ in most cases.

The benefits of introducing parameter α^* when $\alpha^* > 1$ into the WIoU v3 model can be summarized as follows:

1. The model focuses more on high IoU objects by up-weighting their relative loss;
2. Detectors learn faster through up-weighting their relative gradient;
3. Greater flexibility is provided, which allows for achieving different levels of box regression accuracy (i.e., the AP value measured at different IoU thresholds) by adjusting the reweighting scheme of parameter α^*;
4. By adjusting the value of α^*, the detection model can be better adapted to different application scenarios;
5. There is no need to introduce additional parameters to the models.

In addition, this study chooses WIoU v3 as the baseline loss function for the following reasons: first, WIoUv3 integrates the advantages of EIoU and SIoU, conforming to the design idea of exemplary loss function; second, WIoUv3 adopts a dynamic non-monotonic method to evaluate the quality of anchor boxes, paying special attention to instances with higher average quality. WIoUv3 improves the model's ability to precisely locate objects by this enhancement; third, WIoUv3 adaptively adjusts the loss weight of small objects in the UAV image object detection scenario, thus enhancing the effectiveness of model detection.

For α^*-WIoU v3, this study suggests initially setting α and δ to achieve the desired WIoU effect ($\alpha^*=1$). Once the WIoU performs as expected, the value of α^* should be further adjusted. The recommended ranges for α and δ are [1,3] and [1,5], respectively; the value of $\alpha^* = 2.7$ is recommended. If there are sufficient resources available, it is suggested to select an appropriate value for α^* based on different tasks, where the α^* value is selected from the interval [1,3].

3.4.2. Dynamic sparse attention mechanism

In infrared imaging for air vehicles, objects appear relatively smaller in terms of pixel size compared to the complex background. Therefore, to enhance the ability of the detection model to focus on crucial input features, this study introduces a dynamic sparse attention mechanism, which is known as BiFormer, into the backbone network of the Multi-YOLOv8 model. The core idea of the BiFormer is to eliminate irrelevant key-value pairs at the coarse-grained region level and calculate the token-to-token attention in the remaining regions. The structures of the Bi-Level Routing Attention and BiFormer modules are shown in Fig. 12:

In Fig. 12(a), the input feature map X, denoted as $X \in R^{H\times W\times C}$, is initially subdivided into $S \times S$ subregions. Each subregion comprises HW/S^2 feature vectors. X is reconstituted as X^r, which belongs to $R^{S^2\times HW/S^2\times C}$. Subsequently, the feature vectors undergo linear transformations to generate three matrices, Q, K, and V. Attention relations between regions are then obtained by constructing a directed graph to localize the relevant regions of a given region. It is possible to obtain the region level Q^r and K^r. An expression for the adjacency matrix A^r can be obtained in Eq. 23.

$$A^r = Q^r (K^r)^T \tag{23}$$

The expression for the routing index matrix I^r can be obtained (Eq. 24):

$$I^r = topkIndex(A^r) \tag{24}$$

The expression for K^g, V^g can be obtained from I^r (Eq. 25):

$$K^g = gather(K, I^r),\ V^g = gather(V, I^r) \tag{25}$$

Lastly, the gathered K^g and V^g undergo attention processing, and an additional term, Local Context Enhancement $LCE(V)$, is incorporated to yield the output tensor O (Eq. 26).

$$O = Attention(Q, K^g, V^g) + LCE(V) \tag{26}$$

In Fig. 12(b), the DWConv denotes depthwise convolution, the LN represents layer normalization processing, and the MLP refers to a multilayer perceptron.

Considering the trade-off between the computational power of the air vehicle platform and the detection performance, this study aims to use the efficient attention mechanism within the BiFormer module more rationally. In view of that, the C2f module in the eighth layer of the backbone network is replaced by BiFormer module, as illustrated in Fig. 11. After this replacement, the detection model pays more attention to the key information in the input features rather than the background information. The BiFormer block can efficiently identify the key-value pairs with high correlation and perform attention computations. This

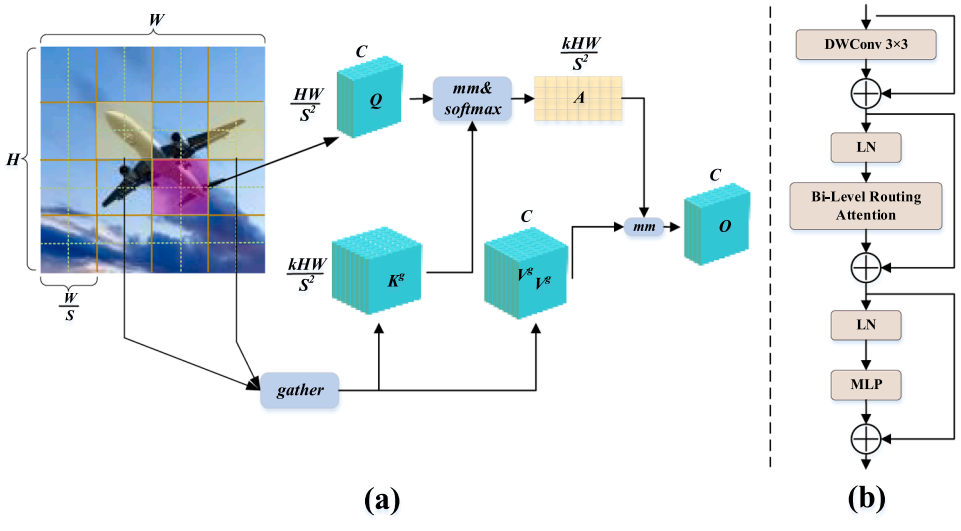


Fig. 12. (a) The Bi-Level Routing Attention structure; (b) The BiFormer module structure.

approach can significantly reduce the computational and storage costs while improving the model's perception of input data.

3.4.3. Lightweight convolution

In pursuit of improved detection accuracy for infrared moving small objects, this study introduces two auxiliary inputs to facilitate detection, albeit at the cost of added time overhead. To strike a balance between model accuracy and speed, the proposed method employs an innovative lightweight convolution technique known as GSConv [57]. The GSConv comprises standard convolution, Depthwise (DW) convolution, and a shuffle module, as illustrated in Fig. 13(a). The structures of the GSBottleneck and GSC2f modules designed based on GSConv are presented in Figs. 13(b) and 13(c), respectively. The GSConv improves the non-linear representation ability of the model by combining DSC layers with shuffle. However, applying GSConv to the entire model may lead to a very deep architecture, which can hinder data flow and increase inference time. Therefore, GSConv is only used in the neck of YOLOv8, which can make the feature map have less redundant information and require no further compression, which allows the attention module to work efficiently.

In the study, the GSC2f modules are used to replace some of the C2f modules in the YOLOv8 model, and the GSConv modules are used to replace some of the Conv modules, as illustrated in Fig. 11.

3.5. Warming-up training method

The proposed Multi-YOLOv8 method uses multiple inputs for object detection. Both primary and auxiliary inputs have certain advantages and disadvantages, as shown in Table 2.

In scenarios where an object is stationary or image registration is inadequate, background suppression might result in a failure of motion spatiotemporal information extraction. Similarly, when an object moves at a very slow speed for a small distance and the background is excessively complex, the extraction of motion spatiotemporal information from the optical flow method might also be unsuccessful.

To prevent potential instability caused by the Multi-YOLOv8 model's over-reliance on the auxiliary inputs, this study proposes a warming-up training method employing a temperature coefficient to regulate the balance between valid and invalid data in the auxiliary inputs. During the training process, the temperature coefficient incrementally increases from zero to one. During the initial training phases, the auxiliary input branches predominantly use invalid data (i.e., pure black images). Once the training iterations surpass a threshold T, the temperature coefficient stabilizes at one. At this stage, the auxiliary inputs exclusively comprise valid data (i.e., background suppression images and optical flow processing images). The training process is illustrated in Fig. 14.

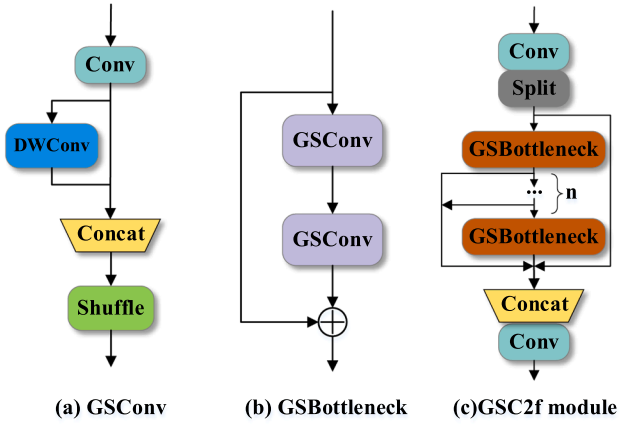


Fig. 13. (a) The structure of GSConv; (b) The structure of GSBottleneck; (c) The structure of GSC2f.

Table 2
The advantages and disadvantages of different types of images.

Image type	Advantages	Disadvantages
Original	I. Rich object information. II. Complete object correlation.	I. Noisy. II. Complex background pixels.
Optical flow	I. Obvious optical flow vector. II. Carrying motion feature.	I. Noisy. II. Small exercise distances are ineffective.
Background suppression	I. Very little background interference. II. Carrying motion feature.	I. Depends on the image registration result. II. Small objects create errors.

According to the properties of the sigmoid function value between zero and one, this study proposes a functional formula for the temperature coefficient T based on the sigmoid function, which is given by Eq. 27.

$$T(E) = 0.02 + 1/\left(1 + e^{4-8*E/E_{threshold}}\right) \tag{27}$$

In Eq. 27, E is the number of rounds (epochs) in the current training, and $E_{threshold}$ is the threshold value that signifies the point where the warming-up phase transitions to the constant temperature phase ($T(E_{threshold}) \approx 1$). Prior to reaching the threshold, the temperature coefficient experiences accelerated growth, and the post-threshold stabilizes at 100 %. The total number of training rounds for the Multi-YOLOv8 model in this study is 200 epochs, and the threshold value is set to 60 epochs. The diagram of the warming-up curve function is presented in Fig. 15.

The warm-up training strategy enables the proposed algorithm to harness the benefits of both single- and multi-input modes. This progressive training approach can effectively mitigate the drawback of an overreliance on the auxiliary inputs in the multi-input mode. Even in scenarios where the auxiliary inputs exhibit suboptimal or failed performance, the algorithm retains the performance baseline of the single-input mode.

For different scenarios, it is recommended to set different values for $E_{threshold}$. Namely, for more complex scenarios, a smaller value is suggested but should be greater than 20; for relatively simple scenarios, a larger value is recommended but should be less than 100.

4. Results and discussion

The proposed Multi-YOLOv8 model is divided into two main phases, the multi-input processing phase and the object detection phase. Accordingly, the experiments were also bifurcated into two primary groups: one for evaluation of the improved optical flow method and another for evaluating the Multi-YOLOv8 object detection model. Section 4.1 presents the dataset used in the experiments. Section 4.2 designs experiments to demonstrate the performance of the PWMHS method in infrared moving small object detection. Section 4.3 designs comparative, ablation, and generalization experiments to demonstrate the effectiveness of the proposed method. Section 4.4 fully discusses the results of the experiment. The hardware platform parameters used include an Intel Core i9–13900 K processor and two NVIDIA RTX 3090Ti GPUs.

4.1. Dataset

The ISD dataset has been designed for the detection and tracking of infrared weak small objects [25]. Data collection primarily focused on aerial fixed-wing unmanned aircraft powered by fuel, with an emphasis on complex backgrounds. This datasets have images with a resolution of 256 × 256 pixels and encompasses a range of complex background scenes, including ground, buildings, mountains, and the sky. It

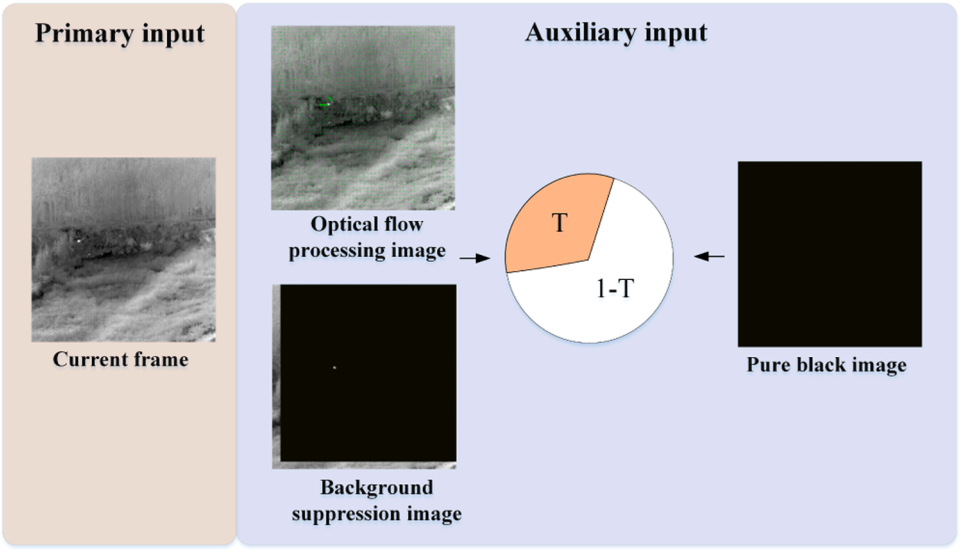


Fig. 14. The schematic diagram of the warm-up training method; T is the temperature coefficient, which is used to regulate the proportion of data input by the auxiliary input branches during the training process. It is important to note that the temperature coefficient T has no effect on the primary input branch.

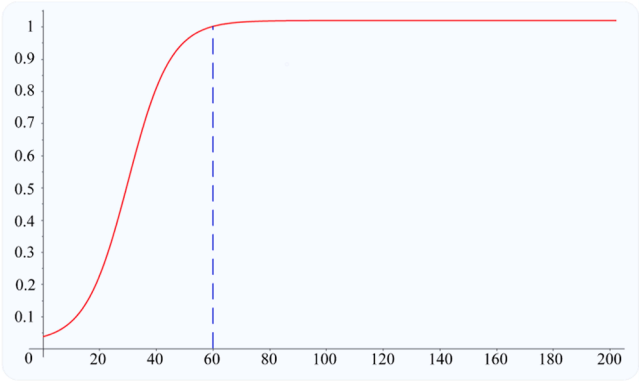


Fig. 15. During the warming-up stage (0–60 epochs), the warming-up curve exhibits gradual increments in the first half (0–30 epochs), starting from a low value, accounting for approximately 4% of the auxiliary inputs. In the second half (30–60 epochs), the proportion of effective data in the auxiliary input branches increase swiftly, aiding the model's transition to a multi-input detection mode. In the constant temperature stage (60–200 epochs), the temperature coefficient remains at 1, and the auxiliary input branches exclusively employ valid data, comprising optical flow processing images and background suppression images.

comprises 22 sequences denoted by data-1– data-22, totaling 16,177 frames and 16,944 individual object instances. Notably, the ISD dataset exclusively pertains to the unmanned aerial vehicle (UAV) category and does not include other object categories.

This study conducted optical flow experiments for each sequence to select optimal model parameters and performed comparative experiments. In the experiments presented in Section 4.3, eight sequences were selected from the 22 sequences as a test set for the ISD based on the balance of detection difficulty and diversity of scenarios, while the remaining sequences constituted the training set. The generalization experiments were performed on the infrared dataset of the Anti-UAV [58] and Anti-UAV410 [59] dataset. The training and test sets were constructed according to the recommended settings in the references. Table 3 provides detailed information about the ISD data division into the training and test sets. Some representative frames in ISD dataset are shown in Fig. 16.

Table 3
Details of training and test set partitioning.

	Category	Sequences
ISD	train	data-2, data-3, data-6, data-7, data-10, data-9, data-11, data-12, data-14, data-16, data-17, data-19, data-20, data-21
	test	data-1, data-4, data-5, data-8, data-13, data-15, data-18, data-22

4.2. Experiments of optical flow method

4.2.1. Experimental design

The study employed three pyramid layers: the zeroth, first, and second layers. The experiments were performed on each dataset sequence in order to select appropriate parameter values. The PWMHS method primarily involved the following parameters: weighting parameters α and k', momentum constant ρ, threshold value T, and optical flow constant λ. After the experiments, it was concluded that $k' = 0.9$ at $\alpha = 30$, $T = 20$, $\rho = 0.75$, and $\lambda = 300$ are the most suitable values, so they were selected for subsequent experiments. The images in Fig. 17 show the visualization results graphs of partial sequences with two iterations at the zeroth layer and seven iterations at the first and second layers for a total of 16 iterations.

4.2.2. Experimental results

While ensuring the accuracy of the optical flow value, this study set the iteration number for both the HS and LK methods to 100, and the iteration number of the PWMHS method was set to 16. The calculated velocity modulus variance (CVMV), relative endpoint error (REPE), and angular error (AE) were selected as evaluation metrics for the optical flow method. The calculation formula was as follows (the values of CVMV, REPE, and AE were the average values of all the frames):

$$CVMV = \frac{1}{2}\left(\sum_{i=1}^{n}(u_{est} - \overline{u_{est}})^2 \Big/ n + \sum_{i=1}^{n}(v_{est} - \overline{v_{est}})^2 \Big/ n\right) \tag{28}$$

$$REPE = \sqrt{\left(u_{est}/u_{gt} - 1\right)^2 + \left(v_{est}/v_{gt} - 1\right)^2} \tag{29}$$

$$AE = \arccos\left(\frac{(u_{est}, v_{est}, 1)^T * (u_{gt}, v_{gt}, 1)}{\sqrt{1.0 + u_{est}{}^* u_{est} + v_{est}{}^* v_{est}}\sqrt{1.0 + u_{gt}{}^* u_{gt} + v_{gt}{}^* v_{gt}}}\right) \tag{30}$$

Where (u_{est}, v_{est}) represents the calculated optical flow value. $(\overline{u_{est}},$

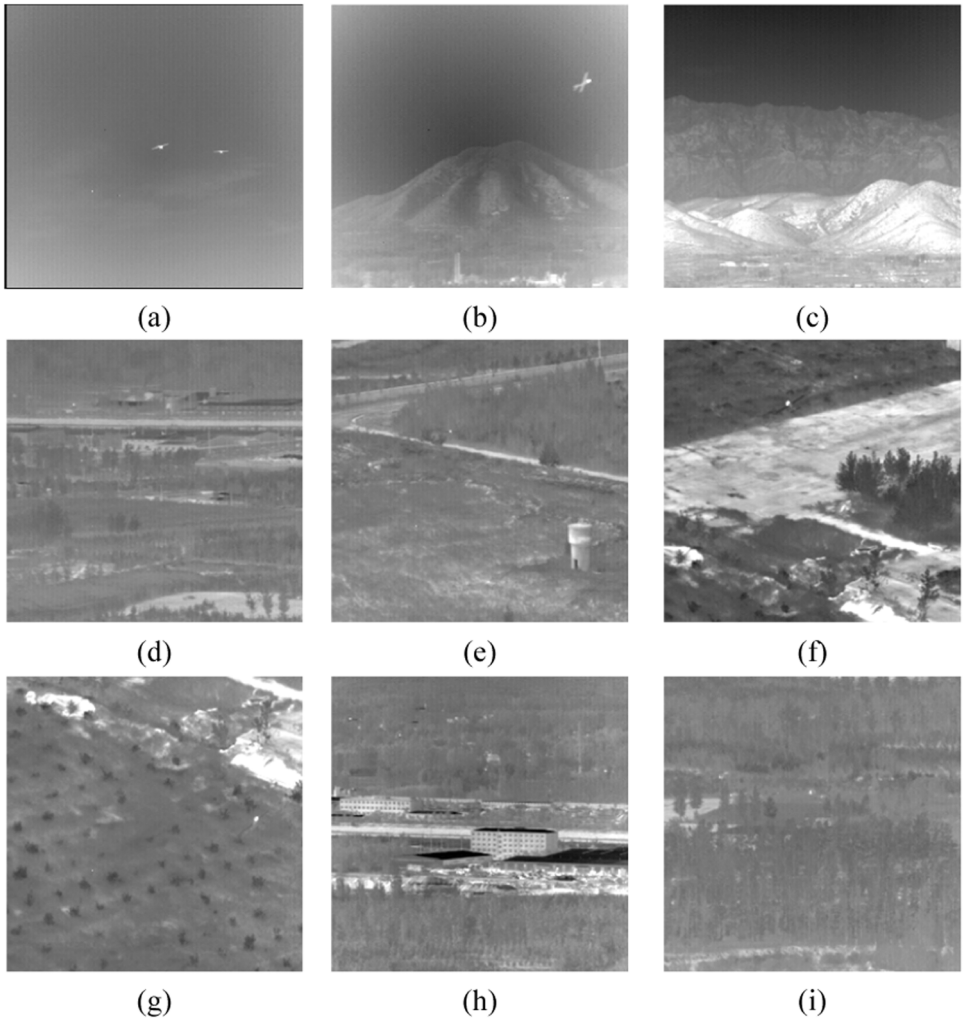


Fig. 16. Some representative frames from the ISD dataset. (a) Empty background, two objects; (b) Simple background, a large object; (c) Simple background, a small object; (d) Complex background, a small object; (e) Complex background, a medium object; (f) Complex background, a large object; (g) Complex background, a large object; (h) Strong background interference, a medium object; (i) Strong background interference, a small object.

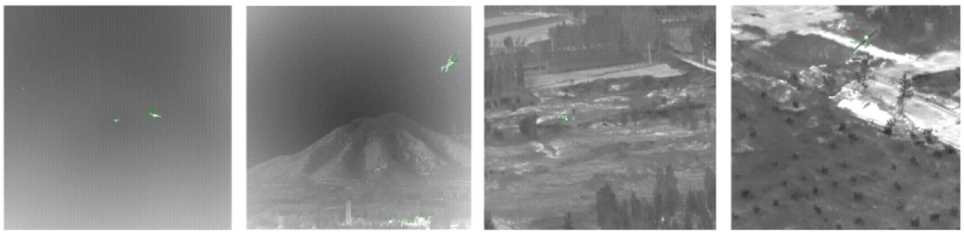


Fig. 17. When α =30, k =0.9, T =20, ρ= 0.75, λ= 300, visualization results for partial sequences of the PWMHS method. The green arrows represent the optical flow vectors. It is evident that the optical flow vectors of the infrared moving small objects can be clearly differentiated from those of the background.

$\overline{v_{est}}$)represents the average of the calculated optical flow values. (u_{gt},v_{gt}) represents the true optical flow value.

The effectiveness of the three metrics lies in providing a detailed assessment of different aspects of optical flow performance, including the stability of velocity modulus, the accuracy of motion endpoint, and the precision of motion direction estimation. By considering these metrics comprehensively, a more holistic understanding of the model's performance in handling optical flow estimation tasks could be obtained. This is significant for evaluating the strengths and weaknesses of optical flow methods.

In Table 4, the results of the optical flow method with different hyperparameters are shown; the AE values are expressed in radians. The calculation results are kept to four decimal places.

After analyzing the experimental results, the following values of hyperparameters were selected: $k' = 0.9$, $\alpha = 30$, $T = 20$, $\rho = 0.75$, and $\lambda = 300$.

Further, after determining the hyperparameters of the PWMHS method through ablation experiments, comparative experiments were conducted for the HS method, the LK method, and the PWMHS method. The experimental results are summarized in Table 5, where the calculation results are retained in four decimal places.

The experimental results in the table showed that the HS method

Table 4
Results of the PWMHS method with different hyperparameters.

Combination	CVMV	REPE	AE
k'= 0.8, α = 20, T=15, ρ=0, λ= 100	0.3562	0.5072	0.2531
k'= 0.8, α = 20, T=20, ρ=0, λ= 100	0.3321	0.5012	0.2510
k'= 0.9, α = 30, T=20, ρ=0, λ= 100	**0.3135**	**0.4998**	**0.2505**
k'= 0.9, α = 30, T=20, ρ=0.15, λ= 100	0.2813	0.4843	0.2412
k'= 0.9, α = 30, T=20, ρ=0.30, λ= 100	0.2614	0.4803	0.2411
k'= 0.9, α = 30, T=20, ρ=0.45, λ= 100	0.2378	0.4747	0.2401
k'= 0.9, α = 30, T=20, ρ=0.60, λ= 100	0.2025	0.4672	0.2380
k'= 0.9, α = 30, T=20, ρ=0.75, λ= 100	**0.1646**	**0.4626**	**0.2313**
k'= 0.9, α = 30, T=20, ρ=0.90, λ= 100	0.1713	0.4674	0.2452
k'= 0.9, α = 30, T=20, ρ=0.75, λ= 200	0.1627	0.4410	0.2311
k'= 0.9, α = 30, T=20, ρ=0.75, λ= 250	0.1614	0.4218	0.2313
k'= 0.9, α = 30, T=20, ρ=0.75, λ= 300	**0.1611**	**0.4001**	**0.2285**
k'= 0.9, α = 30, T=20, ρ=0.75, λ= 350	0.1610	0.4017	0.2287
k'= 0.9, α = 30, T=20, ρ=0.75, λ= 400	0.1608	0.4010	0.2327

Table 5
Results of different optical flow methods. (The top results are highlighted in the table.).

Data-x	Methods	CVMV	REPE	AE
Data-5	HS	0.2997	0.4891	0.3204
	LK	0.2365	0.5933	**0.2040**
	Ours	**0.1412**	**0.3974**	0.2637
Data-8	HS	0.3783	0.4142	0.2474
	LK	0.2735	0.5745	**0.1857**
	Ours	**0.1376**	**0.3226**	0.2732
Data-13	HS	0.3126	**0.4997**	0.2481
	LK	0.4367	0.6864	0.1847
	Ours	**0.1756**	0.5616	**0.1491**
Data-15	HS	0.3470	0.8477	0.4519
	LK	0.3962	0.7148	0.2696
	Ours	**0.1065**	**0.6310**	**0.1937**
Data-18	HS	0.2378	0.4384	0.2982
	LK	0.2045	0.4326	0.1998
	Ours	**0.1376**	**0.3911**	**0.1903**
All	HS	0.3585	0.5201	0.2654
	LK	0.3631	0.5918	0.2574
	Ours	**0.1611**	**0.4001**	**0.2285**

performed better than the LK method in the infrared moving small object detection, which supported the selection of the HS method as the baseline. The proposed method employed a momentum term to accelerate the optical flow calculation, resulting in a highly stable optical flow field. A weighting function was designed, and a threshold T was set to exclude certain points from the iterative calculation, thus reducing computational overhead and suppressing the optical flow calculation for background pixels. In addition, a pyramid structure was used to minimize the number of required iterations. Finally, ablation experiments were conducted to determine the optimal value of the optical flow constant λ. The results indicated that a value within the range of [250,300] yielded the best detection performance. The proposed method obtained a stable vector field while reducing the modulus error and angular error of the optical flow vector. Significantly, in comparison to the traditional HS methods, the proposed method achieved superior optical flow results and faster processing speeds while maintaining a certain level of accuracy and ensuring stability in the velocity field.

The visualization results of the classical HS method with different iteration numbers and the PWMHS method with different parameter settings are shown in Fig. 18.

The results in Fig. 18 show that the HS method achieved relatively good results after approximately 100 iterations when the background was relatively simple, and the object's moving speed was moderate (data-4 and data-20). However, the proposed method outperformed the baseline in terms of optical flow estimation on most datasets after only 10 iterations; namely, after 16 iterations (two iterations in the zeroth layer and seven iterations in the first and second layers), there was a substantial enhancement in the optical flow estimation performance. The PWMHS method could effectively distinguish the object and the background. However, for the optical flow method, it might be challenging to extract object motion features in cases where the object moving distance is short and the contrast with the background is limited. In such situations, combining the background suppression method with the PWMHS method could provide a complementary solution.

4.3. Experiments of Multi-YOLOv8

4.3.1. Experimental design

During model training in this paper, stochastic gradient descent (SGD) with a momentum of 0.937 was employed. The initial learning rate was set at 0.03, and the final learning rate was set at 0.0001. Training was conducted using a batch size of 32 for over 200 epochs, with input images of size 256×256. The weight decay was 0.0005. The mosaic method was the data augmentation strategy, but mosaic data augmentation stopped in the last 10 epochs of training to expedite convergence. The warm-up training method proposed applied to Multi-YOLOv8 during the training process, with the $E_{threshold}$ at 60 epochs. Other comparative algorithms, in contrast, applied the original training method of YOLOv8. The confidence threshold for objects was 0.001, the NMS (Non-Maximum Suppression) IoU threshold was 0.6, and the batch size was 1 during the testing phase.

For the purpose of evaluating the proposed object detection model, this study used several metrics: Precision, Recall, AP_{50} (average precision with the IoU threshold set to 0.5), $AP_{50:95}$ (average precision over the IoU range from 0.5 to 0.95), and FPS (Frames Per Second), and model size.

Precision is defined as the proportion of positive samples predicted by the model relative to all the detected samples, and it is computed as presented in Eq. 31.

$$Precision = \frac{X_{TP}}{X_{TP} + X_{FP}} \tag{31}$$

Recall is defined as the ratio of the number of positive samples correctly predicted by the model to the total number of positive samples that actually appeared. Recall is computed as presented in Eq. 32:

$$Recall = \frac{X_{TP}}{X_{TP} + X_{FN}} \tag{32}$$

where X_{TP} represents cases where a sample is predicted as positive and is indeed a positive sample, X_{FP} indicates instances where a sample is predicted as positive but is actually a negative sample, and X_{FN} denotes cases where a sample is predicted as negative but is actually a positive sample.

The average precision (AP) is equivalent to the area under the precision-recall curve and is computed as demonstrated in Eq. 33:

$$AP = \int_0^1 Precision\left(Recall\right) d\left(Recall\right) \tag{33}$$

Precision indicates whether a model's predictions for the correct category are precise. Recall shows whether the model can capture objects accurately. The AP_{50} metric emphasizes the model's performance at lower IoU thresholds, while $AP_{50:95}$ provides a more comprehensive assessment across a range of IoU thresholds from lower to higher. In the scenario with infrared moving small objects, $AP_{50:95}$ offers a more detailed and comprehensive evaluation than AP_{50}. The four metrics can effectively measure the detection performance of a model.

The YOLOv8s model was used as a baseline model in the experiments. Specifically, the single-YOLOv8 model denoted the YOLOv8s model configured with the original frames as its only input; the together-YOLOv8 model was based on the YOLOv8s model and combined the original frame, optical flow processing image, and background suppression image, stacking them together as the RGB channel input, as

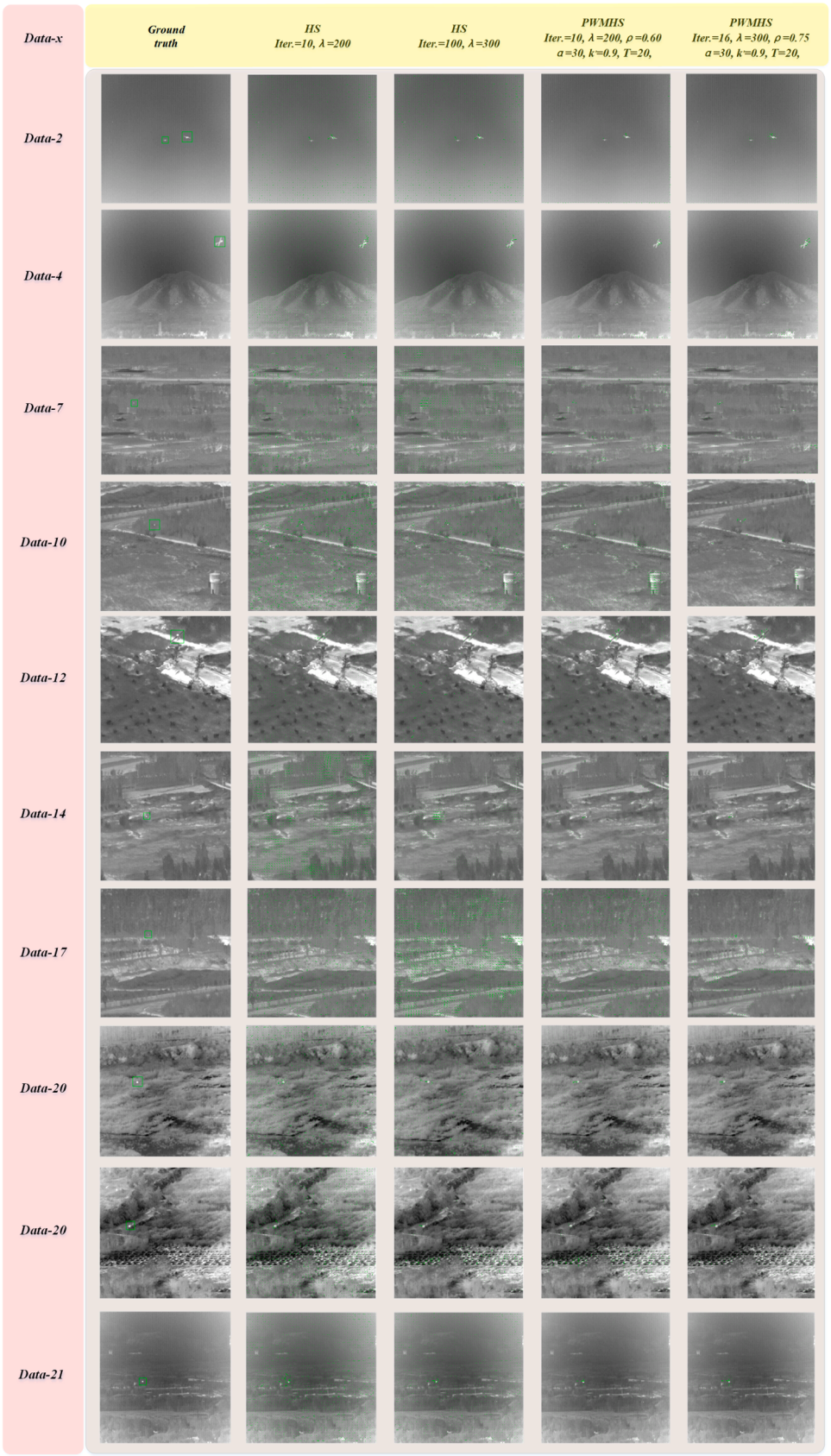


Fig. 18. The visualization results of different methods under different parameters. The images for simple backgrounds (Data-2, Data-3) indicate that both the original HS method and the PWMHS method had good object highlighting ability, but the PWMHS method performed better in suppressing the background pixels. For complex backgrounds (Data-10, Data-14, and Data-20), the HS method failed to localize the object even after 100 iterations and suffered from strong background interference. In contrast, for the proposed method, the optical flow vectors in the object region were significantly different from the background.

shown in Fig. 10. The Multi-YOLOv8 model was the enhanced YOLOv8s model proposed in this study, and it incorporated various improvements into the original YOLOv8s model. It should be noted that the training and parameter settings for all experiments were consistent.

Comparative experiments were performed on the ISD dataset using the YOLOv3, YOLOv5s, YOLOv7-tiny, RFBnet, RefineDet, ISTDet, UIU-Net, Dim2Clear-Net, DNA-Net, single-YOLOv8, together-YOLOv8, and Multi-YOLOv8.

The generalization experiments were performed on the ISD, Anti-UAV dataset, and Anti-UAV410 dataset.

4.3.2. Experiments of improved YOLOv8

Eight sequences were selected from the ISD dataset as a test set (Table 3). The number of images varied significantly across the sequences, with 'data5' having 2997 images and 'data4' having only 396. To balance the detection difficulty of frames in different sequences, 250 frames were randomly chosen from each dataset for testing, resulting in a total of 2000 results for performance evaluation.

4.3.2.1. Comparative experiment of loss function. To obtain more suitable parameter settings, this study conducted experiments on hyperparameters δ,α, and$\alpha^*$of the α*-WIoU v3 function on the YOLOv8s model; the results are shown in Table 6.

Next, to verify the effectiveness of the α*-WIoU v3 improvement, comparative experiments were conducted on the YOLOv8s model using the α*-WIoUv3 and other loss functions under consistent parameters and training conditions.The comparative experimental results are presented in Table 7, where it can be seen that the model performed optimally when the α*-WIoU v3 was employed as a bounding box regression loss.

Compared to the classical loss functions, the proposed loss function exhibited excellent performance in both precision and recall. The AP_{50} and $AP_{50:95}$ were improved by 1.2 % and 5.9 % compared to the CIoU, respectively. In addition, the performance of the WIoU function was improved by introducing hyperparameter α*, which significantly enhanced both AP_{50} (+0.8 %) and $AP_{50:95}$ (+5.6 %) compared to the original WIoU function. This was due to the fact that α* increased the weight of the relative loss, which focused on high IoU objects and improved the model's accuracy in terms of high AP. The experimental results demonstrated that the incorporation of the α*- WIoU v3 function enhanced detection performance.

4.3.2.2. BiFormer block effect. This study introduced the BiFormer module into the YOLOv8 backbone network to enhance essential information detection. Through the experiments, the position of the BiFormer in the YOLOv8 network was determined to ensure that the model could achieve good detection performance. The YOLOv8 model with the α*-WIoUv3 loss function was used as baseline; the results are presented in Table 8.

The "BiFormer-2" indicates the replacement of the C2f module with the BiFormer module, which is positioned at the second layer, as shown in Fig. 11. The results showed that replacing the C2f module in the eighth layer resulted in the best detection performance. The BiFormer module introduced a novel dynamic attention mechanism, enabling the model to emphasize crucial information in input features more effectively. Compared to the baseline model, the AP_{50} and $AP_{50:95}$ were relatively improved by 0.9 % and 0.6 %, respectively. The sparsity of the BiFormer module was achieved by replacing global queries with sparse queries that focused on specific key-value pairs. This design reduced the number of parameters and size of the model while maintaining detection performance.

4.3.3. Comparative experiments

The comparative experiment was conducted to evaluate the performance of the proposed Multi-YOLOv8 and other mainstream models, including YOLOv3, YOLOv5s, YOLOv7-tiny, RFBnet, RefineDet, ISTDet, UIU-Net, Dim2Clear-Net, DNA-Net, YOLOv8 models. These methods were selected for the comparative experiments because they all denote representative or advanced methods in the field of infrared detection. The experimental results are shown in Table 9.

The comparison models are briefly described in the following: the ISTDet model [60] used an image filtering module to obtain a confidence map, which enhanced the response of infrared small objects but suppressed the response of the background. The UIU-Net model [61] employed a nested network to achieve multi-level, multi-scale representational learning of objects. The Dim2Clear-Net model [62] had the ability to refine segmentation results by supplementing details from the original input image. The DNA-Net [63] integrated and used contextual information for small objects through repeated feature fusion and enhancement.

Based on the experimental results presented in Table 9, the early YOLO series algorithms, including YOLOv3, had complex structures and relatively low detection accuracy. Notably, the RFBnet, RefineDet, and ISTDet models that denoted representative object detectors achieved good performance in general object detection. It is important to mention that these three object detection methods were specifically designed for detecting small objects [60]. As the newest algorithm in the YOLO series, the YOLOv8 model has been proven to exhibit higher detection performance compared to the other existing algorithms. This has been the main motivation to use this model as a baseline model in this study. The YOLOv8s has a smaller model size, fewer parameters, and better

Table 6
Results for different values of the hyperparameters of α*-WIoU v3. (The top results are highlighted in the table.).

α*-WIoU v3	Precision	Recall	AP_{50}	$AP_{50:95}$
α=1.4, δ=5.0, α^*=1.0	0.836	0.608	0.671	0.465
α=1.6, δ=4.0, α^*=1.0	0.837	0.611	0.679	0.469
α=1.8, δ=3.0, α^*=1.0	0.837	0.610	0.681	0.471
α=1.9, δ=2.7, α^*=1.0	0.841	0.613	0.684	0.473
***α*=2.0, *δ*=2.5, *α**=1.0**	**0.844**	0.613	0.691	0.474
α=2.2, δ=2.3, α^*=1.0	0.840	0.611	0.682	0.471
α=2.0, δ=2.5, α^*=2.0	0.844	0.613	0.693	0.485
α=2.0, δ=2.5, α^*=2.5	0.843	0.613	0.694	0.497
***α*=2.0, *δ*=2.5, *α**=2.7**	0.843	**0.615**	**0.697**	**0.501**
α=2.0, δ=2.5, α^*=3.0	0.840	0.609	0.692	0.495

Based on the experimental results, the parameters of the α*-WIoU v3 loss function were set to$\alpha = 2$,$\delta = 2.5$ and $\alpha^* = 2.7$ in order to obtain the best detection performance.

Table 7
Comparison of detection results using various loss functions on the YOLOv8s model. (The top results are highlighted in the table.).

Type	Precision	Recall	AP_{50}	$AP_{50:95}$
CIoU	0.835	0.610	0.689	0.473
EIoU	0.827	0.609	0.681	0.472
SIoU	**0.848**	0.612	0.690	0.472
WIoU v1	0.830	0.611	0.688	0.469
WIoU v2	0.834	0.610	0.688	0.471
WIoU v3	0.844	0.613	0.691	0.474
α*-WIOU v3	0.843	**0.615**	**0.697**	**0.501**

Table 8
Results for adding the BiFormer block at different layers within the baseline model's backbone network. (The top results are highlighted in the table.).

Type	Precision	Recall	AP_{50}	$AP_{50:95}$
Baseline	**0.843**	0.615	0.697	0.501
Biformer-2	0.821	0.612	0.693	0.492
Biformer-4	0.833	0.612	0.691	0.501
Biformer-6	0.838	0.617	0.698	0.499
Biformer-8	0.841	**0.618**	**0.703**	**0.504**

Table 9
Results for different methods on the ISD test set. (The top results are highlighted in the table.).

Type	Precision	Recall	AP_{50}	$AP_{50:95}$	FPS
YOLOv3	0.667	0.487	0.521	0.343	59
YOLOv5s	0.823	0.607	0.675	0.452	102
YOLOv7-tiny	0.813	0.606	0.665	0.447	**118**
RFBnet	0.728	0.511	0.572	0.352	68
RefineDet	0.796	0.537	0.636	0.385	40
ISTDet [60]	0.801	0.563	0.656	0.414	69
UIU-Net [61]	0.757	0.549	0.607	0.361	64
Dim2Clear-Net [62]	0.831	0.601	0.678	0.455	49
DNA-Net [63]	0.829	0.604	0.673	0.451	71
Single-YOLOv8	0.835	0.610	0.689	0.473	111
Together-YOLOv8	0.872	0.849	0.889	0.594	68
Multi-YOLOv8	**0.911**	**0.889**	**0.943**	**0.661**	72

detection performance than the other models in the YOLO series. However, in the case of complex backgrounds, the information carried by a single input might not be sufficient to maintain good infrared moving small object detection performance. By efficiently and reasonably expanding the inputs, the object detection model can obtain more comprehensive and effective information to improve the detection performance of infrared moving small objects in complex backgrounds.

In the comparative experiments, the proposed Multi-YOLOv8 demonstrated the highest average detection accuracy and the best overall detection performance among all the models, running in real-time at 72 FPS. Although the improved model had a slightly longer inference time compared to the YOLOv8 model, it still could realize real-time detection, outperforming most detection algorithms. After being accelerated by the TensorRT's FP16 precision mode, the proposed model could achieve faster detection speed. In conclusion, the proposed model demonstrated superior detection performance compared to the other excellent models.

The obtained results can be explained as follows. First, compared with the single-frame detection methods, the multi-frames detection method can capture more motion and object information, and the multi-input processing module can fuse the features of multiple inputs well and save computing power. Second, among the single-frame detection model, the YOLOv8s exhibits the best detection performance, which justifies its selection as a benchmark model in this study. In addition, the proposed α*-WIoU v3 enhances the localization ability and detection performance for high IoU objects in the YOLOv8 model. Moreover, the warming-up training method equips the model with single-input and multi-input object detection capabilities. Further, the BiFormer and GSConv enhance the model's noise suppression and reduce its size, respectively. Finally, the Multi-YOLOv8 maintains good performance when detecting small objects in complex backgrounds, while the performances of the other methods decrease.

4.3.4. Ablation experiments

In this section, to validate the effectiveness of each of the proposed improvement strategies, ablation experiments were performed on the YOLOv8s model. Single-YOLOv8, together-YOLOv8, and Multi-YOLOv8 models were evaluated on eight sequences from the ISD test set. The results are displayed in Table 10, where √ signifies the utilization of the improvement strategy. The positions of the BiFormer, GSConv and GSC2f modules in the Multi-YOLOv8 model are shown in Fig. 11.

The experimental results presented in Table 10 indicate that each of the improvement strategies applied to the baseline model improved the detection performance to a different extent. Specifically, by employing a multi-input processing module with the optical flow processing image, background suppression image, and current frame as inputs, the detection metrics exhibited significant improvement compared to the baseline model. In simple scenarios, using only the primary input for the object detection module saved computational resources and achieved good detection performance. However, in complex scenarios, the auxiliary input helped the primary input complete object detection, improving the model's robustness and detection performance. The multi-input processing module decided whether to activate the auxiliary input processing module based on the detection confidence of the previous two frames, which enabled the effective fusion of features from multiple inputs and ensured the real-time performance.

The proposed α*-WIoUv3 enhanced the model's localization capability and its ability to generalize across different object scenarios, resulting in a 0.8% increase in AP_{50} and a 2.5 % increase in $AP_{50:95}$. The inclusion of the α* parameter enabled the model to focus more on high IoU samples, thus enhancing regression accuracy and the detection performance for high AP.

The integration of the BiFormer module into the backbone network instead of the C2f module increased the model's attention to small objects, while the efficient attention mechanism improved the extraction ability of critical features from the feature maps. In addition, the BiFormer module reduced resource consumption and parameter number by employing sparse queries instead of global queries, leading to a 1.7 % decrease in the model size compared to the YOLOv8 network.

The GSConv integrated lightweight convolution techniques while preserving the Conv operation in the module, which allowed the model to achieve a balance between the lightweight design, improved runtime speed, and preserved detection performance. After replacing certain network components with the lighter GSConv and GSC2f modules, the model size was reduced while the detection performance could still be maintained. This established a foundation for future platform deployment research. Specifically, "21.2 + 16.7" denoted the size of the YOLO network and the auxiliary input processing module, respectively. The GSConv and GSC2f modules reduced the model size by 0.8 MB, which accounted for 2.1 % of the total model size and 3.7 % of the YOLOv8 model size.

The warm-up training strategy provided significant improvements in the recall, AP_{50}, and $AP_{50:95}$ metrics, demonstrating the effectiveness of the warm-up training method. The proportion of auxiliary input images was gradually increased during the training process, which improved the model's balance between single- and multi-input detection capabilities. Furthermore, under the same hardware configuration, inference acceleration was performed using the TensorRT's FP16 precision mode. After accelerating the model using the TensorRT's FP16 precision mode (with a negligible precision loss), the running speeds of the single-YOLOv8, together-YOLOv8, and Multi-YOLOv8 models were 647 FPS, 375 FPS, and 405 FPS, respectively, which met real-time computing

Table 10
Results of the ablation experiments of the Multi-YOLOv8 model. (The top results are highlighted in the table.).

Type	α*-WIoU v3	BiFormer	GSConv & GSC2f	Warming-up training	Prec.	Recall	AP_{50}	$AP_{50:95}$	FPS	Size (MB)
Single-YOLOv8	—	—	—	—	0.835	0.610	0.689	0.473	**111**	22.5
Together-YOLOv8	—	—	—	—	0.872	0.849	0.889	0.594	68	22.5+16.7
Multi-YOLOv8	—	—	—	—	0.923	0.871	0.920	0.629	66	22.5+16.7
Multi-YOLOv8	√	—	—	—	**0.924**	0.874	0.927	0.645	73	22.5+16.7
Multi-YOLOv8	√	√	—	—	0.921	0.879	0.931	0.651	68	22.1+16.7
Multi-YOLOv8	√	√	√	—	0.919	0.877	0.932	0.649	72	21.3+16.7
Multi-YOLOv8	√	√	√	√	0.911	**0.889**	**0.943**	**0.661**	72	**21.3+16.7**

requirements.

The visualization result of object detection is shown in Fig. 19. Each row displays annotations and detections for the same frame. In particular, green boxes represent ground-truth annotations, red boxes are the detected bounding boxes, "uav: 0.88" indicates a UAV object with a confidence score of 0.88, and "Data4_4" denotes that it is the fourth frame of the "data4" sequence. The proposed model demonstrates excellent detection performance even if the small object is dim and the environment is complicated, achieving higher detection confidence and avoiding false positives and misses.

The proposed Multi-YOLOv8 model significantly improved the detection performance by using more comprehensive object information from multiple inputs and the enhanced YOLOv8 model. Compared to the single-YOLOv8 model, the Multi-YOLOv8 model increased in the precision, recall, AP_{50}, and $AP_{50:95}$ values by 7.6 %, 27.9 %, 25.4 %, and 18.8 %, respectively, while maintaining a running speed of 72 FPS that met the real-time requirement for object detection.

4.3.5. Generalization experiments

To validate the generalization capability of the proposed model, this study performed generalization experiments on the ISD, Anti-UAV, and Anti-UAV410 dataset. The experimental settings were the same as those used in the previous experiment. The results are shown in Table 11.

On the three distinct datasets, ISD, Anti-UAV, and Anti-UAV410, a comprehensive analysis of the generalization performance of the Multi-YOLOv8 model was conducted. The ISD dataset included a large number of real images. The infrared portion of the Anti-UAV dataset provided 318 image sequences, including images taken under two lighting conditions (day and night) and different backgrounds (buildings, clouds, trees, etc.). Anti-UAV410 was a dataset for the third Anti-UAV Challenge. Its infrared dataset increased the amount of small objects and complex backgrounds in the dataset. Anti-UAV410 scenes included two different lighting conditions (day and night), two seasons (fall and winter), and various backgrounds such as buildings (30%), mountains (20 %), forests (5 %), urban areas (30 %), clouds (10 %), water surfaces (3 %), etc. The test sets of the Anti-UAV and Anti-UAV410 dataset were used for the generalization experiments.

The results showed that the Multi-YOLOv8 model could maintain high precision and recall on the three datasets, indicating the model's outstanding accuracy in detecting and capturing objects. The model exhibited high AP_{50} and $AP_{50:95}$ values across different datasets and scenarios, suggesting its ability to maintain robust performance when dealing with various object sizes and complex scenes. The Multi-YOLOv8 model demonstrated excellent generalization performance across different datasets and environments, proving its suitability for various infrared moving small object detection tasks.

The visualization results were shown in Fig. 20. The Anti-UAV and Anti-UAV410 datasets included a variety of real-world scenes, including houses, trees, machinery, chimneys, clouds, thick smoke, bridges, roads, mountains, and wheat fields. The Multi-YOLOv8 model used the object feature information and motion information from consecutive frames to accurately detect objects. Even in the situations where the background interference was obvious or the objects overlapped with the background, the model maintained good detection performance (as shown in the 3rd and 4th images of the first row, the 4th and 5th images of the second row, and the 5th and 6th images of the eighth row in Fig. 20). In scenarios featuring small and dark objects, the α*-WIoU v3, BiFormer, and the small object detection layer improved the model's ability to localize, attend to critical information, and detect small objects, respectively, thus achieving excellent detection results (as shown in the 1st and 2nd images of the fifth row and the 3rd and 4th images of the seventh row in Fig. 20). The visualizations indicated that the Multi-YOLOv8 model has excellent detection capabilities for small and dark objects in various complex scenarios, demonstrating outstanding generalization performance on unseen datasets.

Consequently, the results of ablation, comparative, and generalization experiments demonstrated that the proposed Multi-YOLOv8 model outperformed the mainstream infrared moving small object detection algorithms and had good generalization ability. The Multi-YOLOv8 model exhibited exceptional detection performance, maintaining stability even in challenging scenarios with complex backgrounds and small objects. The proposed model can achieve high detection accuracy, stability, and real-time performance.

4.4. Discussion

The proposed Multi-YOLOv8 is an end-to-end infrared moving small object detector based on the YOLO model. It can efficiently handle multiple inputs and use the information they carry for object detection. The experimental results on the ISD, Anti-UAV, and Anti-UAV410 dataset demonstrate the excellent detection performance of the proposed Multi-YOLOv8 for infrared moving small objects. The detection speed of the proposed method can reach almost 72 FPS with an input image resolution of 256×256, which makes the proposed method suitable for real-time infrared moving small object detection. The Multi-YOLOv8 outperforms other object detectors regarding both quantitative and qualitative evaluation metrics. It leverages object and motion information from consecutive frames and incorporates detection heads for various scales, which makes it more adaptable to complex background scenarios in infrared moving small object detection. The proposed loss function α*-WIoU v3, the warming-up training method, the introduced attention mechanism BiFormer, and the lightweight convolution GSConv collectively contribute to the improvement in the proposed model's detection performance. The authors hope that these methods could inspire further research in the field of infrared moving small object detection.

The YOLO-based object detectors rely heavily on datasets. However, the availability of open datasets for infrared small objects is limited, which hinders further development of CNN-based object detectors. Therefore, it is essential to construct more open-access datasets for infrared small object detection. In the future, the authors will conduct in-depth research on efficient and lightweight multi-frame infrared small object detection and tracking methods.

5. Conclusion

In this study, an innovative model for infrared moving small object detection named Multi-YOLOv8 is proposed. The proposed model improves the detection performance by fusing multi-frame information from multiple inputs, and the YOLOv8 network is enhanced to address the challenges of infrared moving small object detection. Based on the results and the above-presented discussion, this study draws several conclusions. First, the improved optical flow method is well-suited for the application scenarios of infrared moving small objects as it can effectively capture object motion information. This is particularly true when optical flow processing images, background suppression images and original frames are combined. The use of multiple inputs, which provide detailed object and motion information, along with the application of multiple-sized detection heads, can significantly enhance detection performance. Second, the multi-input processing module can effectively acquire and utilize information from multiple inputs and integrate the information organically. In this module, the detection confidence values of the previous two frames are used as feedback to decide whether to activate the background suppression module and optical flow processing module or not. Further, the proposed α*-WIoU v3 can improve the model's localization capability and enhance its focus on high IoU objects; particularly, the performance is superior to the CIoU and WIoU in detecting small objects under high AP values. This makes the proposed α*-WIoU v3 a promising option for real-world object detection applications, particularly for small datasets and scenarios with noisy bounding boxes. Moreover, the BiFormer block is introduced as an efficient attention mechanism to enhance the model's focus on

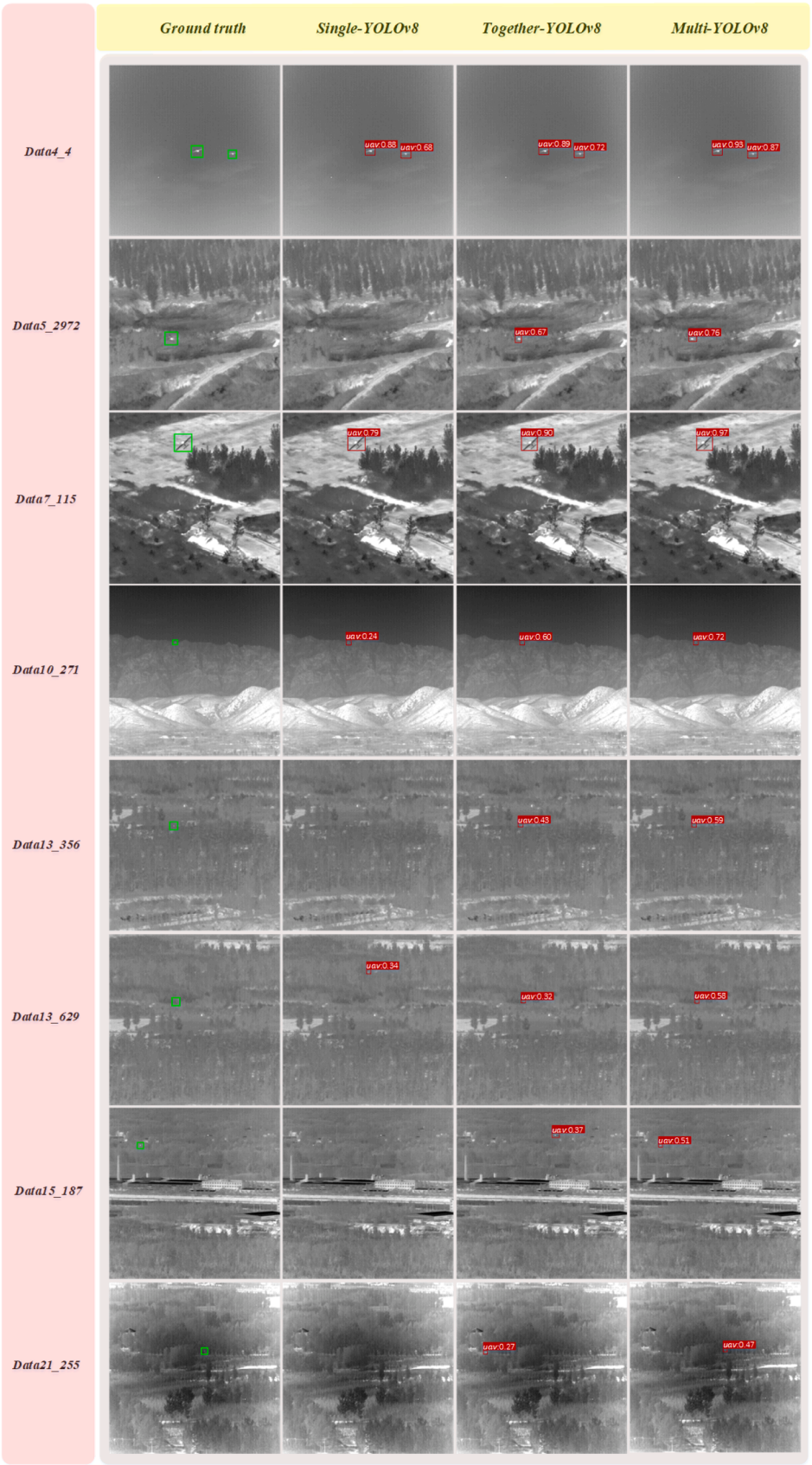


Fig. 19. The visualization results of the object detection obtained on the ISD dataset. For simple backgrounds (Data-4, Data-7), both the Multi-YOLOv8 model and the single-YOLOv8 model could successfully recognize the objects. However, for complex backgrounds (Data-13, Data-15, and Data-21), the single-YOLOv8 model might experience missed and incorrect detections, while the Multi-YOLOv8 model could detect the objects accurately and provide higher detection confidence values.

Table 11
Results of generalization experiments of the Multi-YOLOv8 model on different datasets.

Dataset	Precision	Recall	AP_{50}	$AP_{50:95}$
ISD	0.911	0.889	0.943	0.661
Anti-UAV	0.919	0.891	0.948	0.663
Anti-UAV410	0.892	0.873	0.927	0.646

crucial information. The GSConv and GSC2f modules maintain detection performance while simultaneously increasing the detection speed and reducing the model's size. Furthermore, the warm-up training method provides the model with both multi- and single-input object detection capabilities and increases model stability, thus improving AP_{50} and $AP_{50:95}$ metrics. Finally, compared to the existing detection models, the proposed Multi-YOLOv8 model can achieve better performance regarding detection accuracy and guarantee real-time performance. In

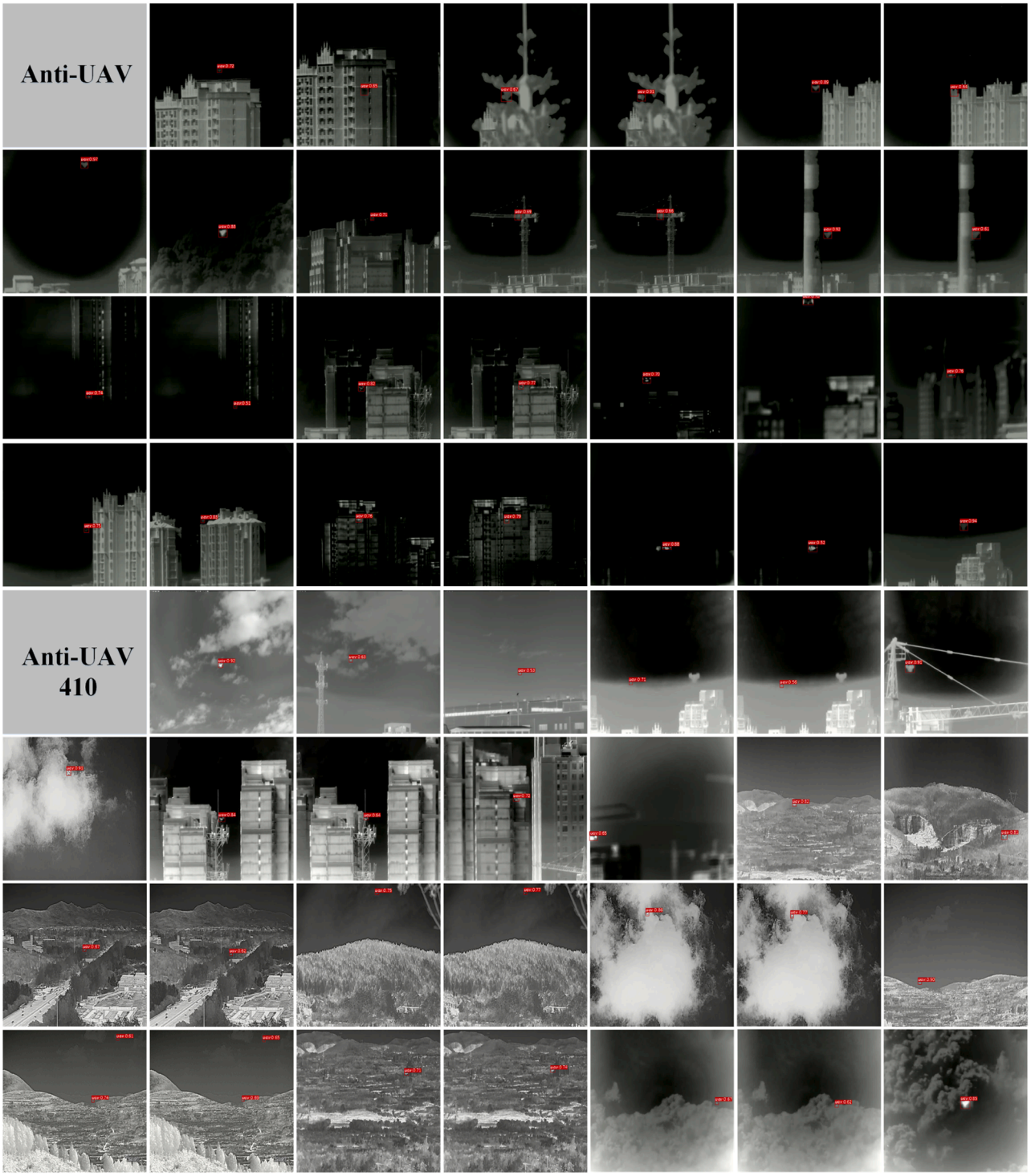


Fig. 20. The visualization results on the Anti-UAV and Anti-UAV410 dataset. These two datasets cover a range of real-world scenarios with varying lighting conditions, backgrounds, and object sizes. The Multi-YOLOv8 model can effectively detect infrared moving small objects in different scenarios while resisting noise, interference, and variation in input data.

summary, embedding multi-frame motion information as auxiliary inputs in the current frame can significantly enhance the effectiveness of input images and the detection capabilities of the model. Future work could consider the application of advanced methods for multi-frame feature fusion and infrared moving small object detection and tracking networks under sandstorm and haze weather conditions.

CRediT authorship contribution statement

Shizun Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Bo Mo:** Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Junwei Xu:** Visualization, Validation, Formal analysis, Conceptualization. **Dawei Li:** Formal analysis, Data curation, Conceptualization. **Jie Zhao:** Resources, Methodology, Conceptualization. **Shuo Han:** Methodology, Funding acquisition, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that has been used is confidential.

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Appendix

The multi-input processing module uses a set of confidence thresholds to adjust the composition of the model inputs. The experimental results under different confidence thresholds are shown in Table I.

Table I
Results for different values of confidence thresholds on the Multi-YOLOv8 model.

Confidence thresholds	Precision	Recall	AP_{50}	$AP_{50:95}$	FPS
(0.15, 0.35)	0.865	0.832	0.881	0.625	77
(0.25, 0.45)	0.901	0.867	0.917	0.639	75
(0.35, 0.50)	**0.911**	**0.889**	**0.943**	**0.661**	**72**
(0.45, 0.55)	0.911	0.890	0.942	0.662	55
(0.50, 0.60)	0.913	0.888	0.944	0.660	49

In the warm-up training method, $E_{threshold}$ is used to regulate the proportion of valid data to pure black images for the auxiliary inputs during training. In this experiment, the setting of $E_{threshold}$ started at 20 epochs, with an interval of 20 epochs. The experimental results of $E_{threshold}$ on the Multi-YOLOv8 model are shown in Table II.

Table II
Results for different values of $E_{threshold}$ with a total of 200 epochs.

$E_{threshold}$	Precision	Recall	AP_{50}	$AP_{50:95}$
20	0.897	0.859	0.920	0.648
40	0.905	0.871	0.929	0.654
60	**0.911**	**0.889**	**0.943**	**0.661**
80	0.887	0.851	0.917	0.641
100	0.867	0.845	0.911	0.637
120	0.849	0.823	0.898	0.621
140	0.839	0.814	0.896	0.615
160	0.837	0.814	0.896	0.611
180	0.819	0.801	0.880	0.601

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